CS 476/676

Numerical Computation Methods
for Financial Models

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Preface and Source Ordering

These notes develop the mathematical and numerical foundations of financial derivatives as taught in CS 476/676. Unlike presentation slides, the exposition gives explicit assumptions, intermediate derivations, and proofs when appropriate. The sequence follows the numbered lecture filenames rather than alphabetical ZIP order. The source archive contains slides for Lectures 1–8 and 10–22; no Lecture 9 slides were supplied. To avoid attributing material to an unprovided lecture, that number is deliberately left vacant. Several lectures contain repeated material as teaching recaps; those recaps are developed where they are most naturally introduced and cross-referenced later.

The core emphasis is numerical valuation: binomial models, Monte Carlo and variance reduction, stochastic differential equations and discretization, quasi-Monte Carlo, jump processes, partial differential equations and finite differences, calibration, American options, and portfolio risk. The accompanying instructor course notes and practice material provide supporting details but are not assigned fictitious lecture numbers.

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Lecture Outline

No.	Subjects developed in the corresponding chapter
1	Derivative contracts, payoff functions, financial risk, risk-free account, arbitrage and replication.
2	One-step binomial tree, hedge equations, pricing probability, call–put parity.
3	Two-step recursion, conditional expectations, recombination, node-dependent hedges.
4	Multi-period lattices, path-dependent claims, American exercise, log-return random walks.
5	Binomial diffusion limit, Brownian motion, Itô calculus, geometric Brownian motion.
6	Normal and lognormal distributions, exact path simulation, Asian calls, MC foundations.
7	Pseudorandom generation, inversion, exponential sampling, MC confidence intervals.
8	Antithetic variates, control variates, optimal coefficients and efficiency.
9	*Not supplied in the source archive; numbering intentionally preserved.*
10	Correlated normals, correlated asset simulation, hedging, Greeks, VaR, ES.
11	Riemann–Stieltjes integration, quadratic variation, construction of Itô integrals.
12	Itô–Doeblin formula, integral examples, martingale and product rules.
13	SDEs, GBM/Vasicek/CIR processes, exact simulation, Euler–Maruyama.
14	Milstein, weak/strong error, Itô–Taylor formulas, multidimensional schemes.
15	Richardson extrapolation, discretization bias, QMC and discrepancy.
16	Poisson processes, jumps, jump diffusion, Black–Scholes PDE and generator.
17	Taylor finite differences, BS grid, explicit scheme, initial/boundary conditions.
18	Monotone/upwind schemes, implicit Euler, Crank–Nicolson, matrix solvers.
19	Stability, convergence, numerical domain, dividend yield and volatility calibration.
20	Optimal stopping, Bermudan recursion, regression/least-squares Monte Carlo.
21	American option bounds, exercise regions, free boundary, complementarity PDE.
22	American-option LCP schemes, Markowitz optimization, simulated VaR and ES optimization.

Lecture 1

Financial Markets, Derivatives, and Discounting

1.1 Financial assets and contingent claims

A financial model consists of an information structure, a collection of traded assets, and rules for constructing self-financing portfolios. Throughout, S_t denotes the price of one unit of a risky asset at time t, while B_t denotes the value of a cash account. The fundamental problem of derivative pricing is to determine the time-t value of a contingent cash flow whose magnitude depends on future market outcomes. The valuation depends on what risks can be traded and hedged, not merely on an investor's subjective prediction of future prices.

Definition 1.1 (European options)**.** A European call with strike $K > 0$ and maturity T grants the holder the right, but not the obligation, to buy the underlying at time T for K. Its terminal payoff is $(S_T - K)^+ = \max\{S_T - K, 0\}$. A European put grants the right to sell for K and has terminal payoff $(K - S_T)^+$. These payoffs depend only on the terminal value and hence are examples of European-style contingent claims.

Example 1.2 (A motivating call)**.** If $S_0 = 100$, $K = 105$, and $T = 1$ year, the call pays zero when $S_T \leq 105$ and pays $S_T - 105$ when $S_T > 105$. The initial call premium is distinct from the eventual payoff. For instance, a realized payoff of 20 does not mean an initial premium of 20 was fair; financing, trading opportunities, and uncertainty must be considered.

The following basic identities expose the piecewise-linear geometry of call and put payoffs:

$$\begin{aligned}(S_T - K)^+ &= (S_T - K)\mathbf{1}_{\{S_T > K\}},\\ (K - S_T)^+ &= (K - S_T)\mathbf{1}_{\{S_T < K\}},\\ (S_T - K)^+ - (K - S_T)^+ &= S_T - K,\\ (S_T - K)^+ + (K - S_T)^+ &= |S_T - K|\,.\end{aligned}$$

A portfolio of options adds the corresponding payoffs. For instance, simultaneously holding a call and a put at the same strike creates a *straddle*, paying $|S_T - K|$, whereas buying the call and selling the put produces the forward-like payoff $S_T - K$.

1.2 Path dependence and the computational problem

Not every payoff depends solely on S_T. For observation dates $0 < t_1 < \cdots < t_m = T$, an arithmetic-average Asian call pays

$$H(S_{t_1}, \ldots, S_{t_m}) = \left(\frac{1}{m}\sum_{j=1}^{m} S_{t_j} - K\right)^+ .$$

A lookback put may pay $\max_{0\le j\le m} S_{t_j} - S_T$. Such payoffs depend on a functional of an entire trajectory. In many models a simple closed-form valuation is unavailable, motivating tree-based recursion, simulation, and PDE methods. A *model price* must also be distinguished from a market quote: the model is useful only to the extent its assumptions and numerical approximations are appropriate.

1.3 Risk-free investment and time value of money

Definition 1.3 (Money-market account)**.** With a constant continuously compounded risk-free rate r, one unit invested at time zero grows to $B_t = e^{rt}$. The time-t value of one unit due at T is $e^{-r(T-t)}$.

For a quoted periodic rate r_m compounded m times per year, the account value at integer year horizons is $(1 + r_m/m)^{mt}$. Since $\log(1 + x) = x + O(x^2)$ as $x \to 0$,

$$\begin{aligned}\log\left(1+\frac{r}{m}\right)^{mt} &= mt\log\left(1+\frac{r}{m}\right)\\ &= mt\left(\frac{r}{m}-\frac{r^2}{2m^2}+O(m^{-3})\right)\\ &= rt-\frac{r^2t}{2m}+O(m^{-2}) \longrightarrow rt.\end{aligned}$$

Exponentiating yields continuous compounding: $\lim_{m\to\infty}(1 + r/m)^{mt} = e^{rt}$. The value at time t of a deterministic amount X paid at T is $Xe^{-r(T-t)}$. A risky cash flow, by contrast, cannot generally be valued by replacing X with its expectation under the real-world probability law.

1.4 Arbitrage, replication, and the modeling ingredients

Definition 1.4 (Arbitrage)**.** An arbitrage is a self-financing trading strategy with zero or negative initial cost, nonnegative terminal wealth in every state, and strictly positive terminal wealth in at least one state of positive probability (with the usual admissibility assumptions in continuous time).

Proposition 1.5 (Law of one price)**.** *If two attainable portfolios have identical terminal payoffs in every state, then absence of arbitrage requires their initial values to coincide.*

Proof. If two identical-payoff portfolios cost $v_1 < v_2$, buy the cheaper and short the more expensive. The initial gain $v_2 - v_1 > 0$ can be deposited in the cash account, while the two terminal portfolio payoffs cancel exactly. This produces an arbitrage, a contradiction. □

Thus, the basic task is to find either a replicating strategy or an equivalent risk-neutral probability distribution under which discounted asset prices are martingales. These two approaches will coincide in a complete binomial market.

1.5 Option portfolios and nonlinear exposure

The payoff diagram of a derivative is informative because it reveals which scenarios create gains and losses. A long call $C_T = (S_T - K)^+$ has nonnegative terminal payout but costs a positive

premium at inception; the total investment profit at maturity, if the premium is financed at rate r, is $(S_T - K)^+ - C_0 e^{rT}$. Thus a positive payoff does not automatically imply a positive net investment return. A short call receives the initial premium but has a potentially unbounded loss as $S_T \to \infty$.

Example 1.6 (Spreads and digital limits)**.** Consider a call spread consisting of a long call at strike K_1 and a short call at $K_2 > K_1$. Its terminal payoff is

$$H(s) = (s - K_1)^+ - (s - K_2)^+ = \begin{cases} 0, & s \le K_1, \\ s - K_1, & K_1 < s < K_2, \\ K_2 - K_1, & s \ge K_2. \end{cases}$$

The payoff is bounded above by $K_2 - K_1$. For $K_2 = K_1 + \varepsilon$, the normalized call spread $H(s)/\varepsilon$ approaches $\mathbf{1}_{\{s>K_1\}}$ for $s \neq K_1$, connecting ordinary calls with digital (binary) option payoffs. Such limiting arguments help explain why the strike derivatives of call prices convey information about risk-neutral distributions.

Remark 1.7 (Price versus risk). The curvature of an option value as a function of its underlying is a measure of nonlinear exposure. The kink at K in the terminal call payoff introduces a nonsmooth derivative at maturity. Even if a smooth pricing formula exists for positive time to maturity, finite-difference approximations of its derivatives may become unstable or require finer meshes close to expiry and strike.

1.6 Derivatives and distributional expectations

For a payoff $H(S_T)$, an elementary but incorrect pricing suggestion is to discount $\mathbb{E}^{\mathbb{P}}[H(S_T)]$ using the stock's historical drift μ. This is not generally a replication cost. In a complete arbitrage-free model, one instead constructs a probability measure $\mathbb{Q}$ such that the discounted prices of traded, properly dividend-adjusted assets are martingales. Under that measure, the fair price is a discounted expectation. The reason is *no arbitrage*, not a hypothesis that agents actually believe the stock's expected return equals r.

For a European claim with payoff H, a risk-neutral valuation functional $\pi(H) = e^{-rT}\mathbb{E}^{\mathbb{Q}}[H]$ is linear. Hence

$$\begin{aligned} \pi(aH_1 + bH_2) &= e^{-rT}\mathbb{E}^{\mathbb{Q}}[aH_1 + bH_2] \\ &= a\pi(H_1) + b\pi(H_2). \end{aligned}$$

Linearity is the mathematical basis for combining derivatives into portfolios and valuing the components separately. It relies on frictionless trading and fixed linear contract cash flows; transaction costs, credit constraints, and nonlinear funding rules can destroy this simple structure.

Lecture 2

One-Period Binomial Pricing and Put–Call Parity

2.1 A one-period stock model

Fix a time interval of length $h > 0$. At time zero the stock price is $S_0 > 0$. At time h, the stock takes one of two values:

$$S_h = \begin{cases} uS_0, & \text{up state,} \\ dS_0, & \text{down state,} \end{cases} \qquad 0 < d < u.$$

The cash account grows by e^{rh}. If the physical probability of an upward move is p, then $\mathbb{P}(S_h = uS_0) = p$. No-arbitrage valuations below will not require knowing p.

Proposition 2.1 (No-arbitrage condition)**.** *For the one-step binomial model with unrestricted trading and borrowing, absence of arbitrage is equivalent to*

$$d < e^{rh} < u.$$

Proof. If $d \geq e^{rh}$, buying one stock funded by borrowing S_0 in cash gives a nonnegative profit in both states and a strictly positive profit in at least one state, unless the model is degenerate. If $u \leq e^{rh}$, shorting stock and lending the proceeds yields the reverse arbitrage. Conversely, when $d < e^{rh} < u$, the risk-neutral probability defined below lies strictly between zero and one and every contingent payoff can be replicated, excluding arbitrage. □

2.2 Replication by stock and cash

Suppose a European payoff equals H_u in the up state and H_d in the down state. Seek a portfolio with Δ shares and β units of the money-market account, normalized so that $B_0 = 1$. The replication equations are

$$\begin{aligned} \Delta u S_0 + \beta e^{rh} &= H_u, \\ \Delta d S_0 + \beta e^{rh} &= H_d. \end{aligned}$$

Subtracting and substituting gives

$$\begin{aligned} \Delta &= \frac{H_u - H_d}{S_0(u-d)}, \\ \beta &= e^{-rh}\frac{uH_d - dH_u}{u-d}, \\ V_0 &= \Delta S_0 + \beta. \end{aligned}$$

Remark 2.2. The hedging ratio Δ measures how many shares the replicating portfolio holds. Its sign and magnitude are determined by the *difference* between the payoffs in the two states. This is the discrete ancestor of continuous-time delta hedging.

2.3 Risk-neutral probabilities

Rearranging the replication cost produces

$$\begin{aligned} V_0 &= \frac{H_u - H_d}{u-d} + e^{-rh}\frac{uH_d - dH_u}{u-d} \\ &= e^{-rh}\left[\frac{e^{rh}-d}{u-d}H_u + \frac{u-e^{rh}}{u-d}H_d\right]. \end{aligned}$$

Define the risk-neutral up probability by $q = (e^{rh}-d)/(u-d)$ and the down probability by $1-q = (u-e^{rh})/(u-d)$. Both are positive under the no-arbitrage condition and $q+(1-q)=1$. Thus

$$V_0 = e^{-rh}\mathbb{E}^{\mathbb{Q}}[H(S_h)].$$

Proposition 2.3 (Discounted stock martingale)**.** *Under* $\mathbb{Q}$, $\mathbb{E}^{\mathbb{Q}}[e^{-rh}S_h] = S_0$.

Proof.

$$\begin{aligned} e^{-rh}\mathbb{E}^{\mathbb{Q}}[S_h] &= e^{-rh}S_0\left[qu + (1-q)d\right] \\ &= e^{-rh}S_0\left[d + \frac{e^{rh}-d}{u-d}(u-d)\right] = S_0. \end{aligned}$$

□

It is crucial that q is a *pricing* probability, not a claim about the empirical frequency of up moves. It is obtained from tradable replication constraints, not statistical estimation of p.

2.4 Put–call parity

Theorem 2.4 (European put–call parity)**.** *For European call and put contracts on the same non-dividend-paying underlying, with matching strike K and maturity T, the initial values satisfy*

$$C_0 - P_0 = S_0 - Ke^{-rT}.$$

Proof. A long call and short put has terminal payoff

$$(S_T - K)^+ - (K - S_T)^+ = S_T - K.$$

A portfolio of one share of stock and a short position in Ke^{-rT} units of today's cash also pays $S_T - K$ at maturity. By the law of one price, $C_0 - P_0 = S_0 - Ke^{-rT}$. □

A parity violation would allow a static arbitrage. Under a deterministic continuous dividend yield a, stock replaced by prepaid forward exposure changes the right-hand side to $S_0e^{-aT} - Ke^{-rT}$.

2.5 A fully worked one-period replication

Let $S_0 = 50$, $u = 1.3$, $d = 0.8$, $e^{rh} = 1.05$, and consider a call with strike $K = 52$. The possible stock outcomes are 65 and 40, and the option payouts are 13 and 0. We first calculate

$$q = \frac{1.05 - 0.8}{1.3 - 0.8} = \frac{1}{2},$$
$$C_0 = e^{-rh}\left(\frac{1}{2} \cdot 13 + \frac{1}{2} \cdot 0\right) = \frac{6.5}{1.05}.$$

To derive the corresponding hedge without invoking expected values, solve the two-equation replication system:

$$\begin{aligned} 65\Delta + 1.05\beta &= 13, \\ 40\Delta + 1.05\beta &= 0, \\ 25\Delta &= 13, \\ \Delta &= \frac{13}{25} = 0.52, \\ \beta &= -\frac{40(0.52)}{1.05} = -\frac{20.8}{1.05}. \end{aligned}$$

The initial portfolio cost is

$$50\Delta + \beta = 26 - \frac{20.8}{1.05} = \frac{6.5}{1.05},$$

exactly equal to the risk-neutral expected discounted payoff. If the market sold the same call for a smaller amount, we could buy the call and short this replicating portfolio; if it sold the call for more, we could short the call and buy the replicating portfolio. Either discrepancy creates a state-independent net cash inflow at time zero without future obligations.

2.6 Uniqueness and completeness in a two-state market

The two replication equations form a linear system

$$\begin{pmatrix} uS_0 & e^{rh} \\ dS_0 & e^{rh} \end{pmatrix} \begin{pmatrix} \Delta \\ \beta \end{pmatrix} = \begin{pmatrix} H_u \\ H_d \end{pmatrix}.$$

Its determinant is $S_0 e^{rh}(u - d) \neq 0$ when $u > d$. Therefore every vector of state-contingent cash flows has a unique representation as a stock-and-bond portfolio. This is exactly what *market completeness* means in the one-period model. Completeness explains why the no-arbitrage price is unique: not only can prices be bounded by arbitrage arguments, but every payoff is attainable by trading existing securities.

The corresponding risk-neutral probability vector can be derived by imposing the two pricing conditions for the basis assets. If $\mathbb{Q}(u) = q$ and $\mathbb{Q}(d) = 1 - q$, then discounting the cash account must reproduce its current value and discounting the stock must reproduce S_0. The cash constraint is $q + (1 - q) = 1$, and the stock constraint is $e^{-rh}(qu + (1 - q)d) = 1$. The latter uniquely determines q. This algebra shows that replication and the equivalent martingale measure are two descriptions of the same linear system.

Lecture 3

Two-Period Trees and Dynamic Risk-Neutral Pricing

3.1 Dynamic replication

Extend the one-period model to two dates $0, h, 2h$ with independent binomial up/down innovations under $\mathbb{Q}$. If $S_h = s$, let the remaining payoff in the next step equal $H(us)$ or $H(ds)$. The conditional value at h must then be

$$V_h(s) = e^{-rh}\big[qH(us) + (1-q)H(ds)\big].$$

At time zero, applying the same replication principle once again yields

$$\begin{aligned} V_0 &= e^{-rh}\left[qV_h(uS_0) + (1-q)V_h(dS_0)\right] \\ &= e^{-2rh}\left[q^2H(u^2S_0) + 2q(1-q)H(udS_0) + (1-q)^2H(d^2S_0)\right], \end{aligned}$$

when the payoff depends only on the terminal stock price. The factor two counts the distinct up–down and down–up paths that reach the same node in a recombining tree.

3.2 Conditional expectations and the tower property

Let $\mathcal{F}_n$ be the information available by date $t_n = nh$. For a claim H payable at $T = Nh$, risk-neutral pricing at intermediate dates takes the form

$$V_{t_n} = \mathbb{E}^{\mathbb{Q}}\big[e^{-r(T-t_n)}H \mid \mathcal{F}_n\big].$$

For $n < N$ the tower property implies

$$\begin{aligned} V_{t_n} &= e^{-r(T-t_n)}\mathbb{E}^{\mathbb{Q}}\left[\mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n\right] \\ &= e^{-rh}\mathbb{E}^{\mathbb{Q}}[V_{t_{n+1}} \mid \mathcal{F}_n]. \end{aligned}$$

This is the mathematical basis for backward induction. It says that the *discounted value process* is a martingale under $\mathbb{Q}$, while the undiscounted option value need not be a martingale.

3.3 Node-by-node hedge ratios

A fixed hedge placed at inception is generally not sufficient after future information is revealed. At node (n, s), let the two next option values be V^u and V^d. The locally replicating positions are

$$\Delta_n(s) = \frac{V^u - V^d}{s(u-d)},$$

$$\beta_n(s) = e^{-rh}\frac{uV^d - dV^u}{u-d}.$$

The agent rebalances these positions as the path evolves. The strategy is self-financing when trading costs and cash flows satisfy the rebalancing identity: stock purchases must be financed by reductions in the cash position. Completeness of the binary tree follows because two traded assets span the two possible next-step payoffs at every node.

Example 3.1 (A European call in two periods)**.** Take $S_0 = 100$, $u = 1.2$, $d = 0.8$, $e^{rh} = 1.02$, and $K = 100$. The pricing probability is $q = (1.02 - 0.8)/(1.2 - 0.8) = 0.55$. Terminal stock values are $144, 96, 64$, corresponding to call payoffs $44, 0, 0$. Hence

$$C_0 = e^{-2rh}\left(0.55^2 \cdot 44\right) = \frac{13.31}{1.02^2}.$$

A physical up probability different from 0.55 changes the distribution of realized returns but not this complete-market replication price.

3.4 Induction proof of multi-period pricing

Theorem 3.2 (Backward valuation in a complete finite tree)**.** *Suppose each one-step stock tree obeys $d < e^{rh} < u$, trading is frictionless and self-financing, and the claim pays an integrable $\mathcal{F}_N$-measurable amount H at maturity. Then its arbitrage-free value at time n is*

$$V_n = e^{-rh}\mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_n], \qquad V_N = H,$$

and hence $V_n = e^{-r(N-n)h}\mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_n]$.

Proof. At time $N-1$, the claim has only two possible payoffs conditional on the current state. The one-step replication argument establishes the discounted conditional expectation formula. Suppose the value V_{n+1} is attainable by trading over dates $n+1$ through N. At time n, replicate the two next-step values V^u_{n+1}, V^d_{n+1} using a stock/cash portfolio over one period, then switch to the already known continuation replicating strategies after the next state is observed. Concatenating these strategies yields a self-financing replication of H from time n. Its time-n cost is the one-step discounted conditional expectation. The tower property then establishes the full discounted expectation. Backward induction completes the proof. □

3.5 Nonrecombining state variables and path counts

A recombining stock tree with N time steps has only $N+1$ possible terminal *stock values*, but it still has 2^N distinct sequences of up and down moves. A European claim $H(S_N)$ assigns the same payoff to all paths ending at the same stock price; aggregation by binomial multiplicities is legitimate. By contrast, for a lookback option $H = (M_N - S_N)$ with $M_N = \max_{0 \le k \le N} S_k$, two paths with the same terminal stock price may have distinct past maxima. The state of a valid dynamic program must distinguish them.

Example 3.3 (A four-step path distinction)**.** Take $u = 2$ and $d = 1/2$, starting from $S_0 = 1$. The paths $UUDD$ and $UDUD$ both finish at $S_4 = 1$, but their running maxima are 4 and 2. Their lookback-put payouts are therefore 3 and 1. A recursion using S_n alone would incorrectly merge these claims and misprice the option. Enlarging the state to (S_n, M_n) yields the correct transition

$$(s, m) \longmapsto (us, \max\{m, us\}) \quad \text{or} \quad (ds, \max\{m, ds\}).$$

Lecture 4

Multi-Period Lattices, American Options, and Random Walks

4.1 Multi-period stock dynamics

Let $\xi_1, \dots, \xi_N$ be up/down multipliers equal to u or d. The price process is

$$S_n = S_0 \prod_{j=1}^{n} \xi_j, \quad S_{n+1} \in \{uS_n, dS_n\}.$$

If $d = u^{-1}$, the possible terminal prices are $S_N = S_0 u^{2k-N}$ for $k = 0, \dots, N$. If the multipliers are constant, the tree recombines because $ud = du$. A European terminal claim $H(S_N)$ has value

$$V_0 = e^{-rNh} \sum_{k=0}^{N} \binom{N}{k} q^k (1-q)^{N-k} H(S_0 u^k d^{N-k}).$$

This formula is useful for path-independent payoffs. For path-dependent claims, distinct paths reaching the same stock price can have different payoff-relevant histories, so they cannot always be merged into a single state.

Algorithm 4.1 (Backward induction for a European claim)**.** Initialize the terminal node values $V_{N,k} = H(S_0 u^k d^{N-k})$. For $n = N-1, \dots, 0$ and $k = 0, \dots, n$, compute

$$V_{n,k} = e^{-rh}\left[qV_{n+1,k+1} + (1-q)V_{n+1,k}\right].$$

The time-zero price is $V_{0,0}$. Reconstruct hedge ratios at each node using the two successor values.

4.2 Path-dependent and early-exercise claims

For an arithmetic Asian claim, a natural enlarged state variable is (S_n, A_n), where $A_n = \sum_{j=1}^{n} S_j$. The transition satisfies $A_{n+1} = A_n + S_{n+1}$. For a lookback option, the running maximum $M_n = \max_{0 \le j \le n} S_j$ serves as an additional state. The backward pricing function is therefore $V_n(s, m)$ or $V_n(s, a)$ rather than $V_n(s)$ alone.

Definition 4.2 (American-style early exercise)**.** An American derivative with intrinsic exercise value $g(s)$ may be exercised at any permitted date. On a discrete exercise grid it satisfies

$$V_{n,k} = \max\left\{g(S_{n,k}),\ e^{-rh}\left[qV_{n+1,k+1} + (1-q)V_{n+1,k}\right]\right\}.$$

The second entry is the continuation value, whereas the first is the gain from exercising now. The recursion optimizes over stopping decisions that use only available information, anticipating the optimal stopping and complementary problems developed later.

4.3 Log returns as a random walk

Let $Y_n = \log(S_n/S_0)$. Then

$$Y_n = \sum_{j=1}^{n} Z_j, \qquad Z_j = \begin{cases} \log u, & \xi_j = u, \\ \log d, & \xi_j = d. \end{cases}$$

If $d = u^{-1}$, write $a = \log u$. Then $Z_j \in \{-a, a\}$ and

$$\begin{aligned} \mathbb{E}[Z_j] &= a(2p-1), \\ \mathrm{Var}(Z_j) &= \mathbb{E}[Z_j^2] - \mathbb{E}[Z_j]^2 = a^2 - a^2(2p-1)^2 = 4a^2p(1-p). \end{aligned}$$

By independence, $\mathbb{E}[Y_n] = na(2p-1)$ and $\mathrm{Var}(Y_n) = 4na^2p(1-p)$. The law under $\mathbb{P}$ uses p; under $\mathbb{Q}$ the same identities use q. A diffusive scaling of these log returns leads toward Brownian motion and geometric Brownian motion.

4.4 Optimal stopping on a lattice: a proof

Proposition 4.3 (Bermudan dynamic programming)**.** *Suppose a claim offers reward $G_n = g(S_n)$ upon exercise at date n. Set $V_N = G_N$ and, for $n < N$, define*

$$V_n = \max\{G_n, e^{-rh}\mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_n]\}.$$

Then V_n is the essential supremum of conditional discounted exercise payoffs over all stopping times $\tau \in \{n, \dots, N\}$, and a stopping time that first enters the exercise set is optimal.

Proof. At maturity there is no choice, so $V_N = G_N$. Suppose the assertion holds starting from $n+1$. At date n, an admissible strategy either exercises immediately for G_n, or waits one period. In the latter case, the best attainable continuation value is, by the induction hypothesis, $e^{-rh}\mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_n]$. Taking the larger of the two values captures all admissible decisions, and the rule choosing the maximizing action is adapted. Backward induction proves optimality. □

4.5 Binomial convergence to a diffusion

In the symmetric-multiplier scheme $u = e^{\sigma\sqrt{h}}$, $d = e^{-\sigma\sqrt{h}}$, write the log increment as $\Delta \log S = \sigma\sqrt{h}\,\varepsilon$ where $\varepsilon \in \{-1, 1\}$. Its risk-neutral mean and variance can be expanded using $q_h = 1/2 + (r - \sigma^2/2)\sqrt{h}/(2\sigma) + O(h)$:

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}}[\Delta \log S] &= \sigma\sqrt{h}(2q_h - 1) \\ &= (r - \sigma^2/2)h + O(h^{3/2}), \\ \mathrm{Var}^{\mathbb{Q}}(\Delta \log S) &= \sigma^2 h - \left(\mathbb{E}^{\mathbb{Q}}[\Delta \log S]\right)^2 \\ &= \sigma^2 h + O(h^2). \end{aligned}$$

Thus the aggregate log return over $N = T/h$ steps has limiting mean $(r - \sigma^2/2)T$ and variance $\sigma^2 T$. A functional central limit theorem can strengthen this marginal statement to convergence of the entire rescaled price process. The resulting stock process is geometric Brownian motion, whose exponential form guarantees positive prices.

Lecture 5

From Binomial Models to Itô Processes

5.1 The diffusion limit

Consider an interval of duration h with symmetric log-price increments of order $\sqrt{h}$. The Cox–Ross–Rubinstein multipliers are

$$u = e^{\sigma\sqrt{h}}, \qquad d = e^{-\sigma\sqrt{h}}, \qquad q_h = \frac{e^{rh} - d}{u - d}.$$

Expand around $h = 0$ to see the risk-neutral drift:

$$\begin{aligned} u &= 1 + \sigma\sqrt{h} + \tfrac{1}{2}\sigma^2 h + O(h^{3/2}), \\ d &= 1 - \sigma\sqrt{h} + \tfrac{1}{2}\sigma^2 h + O(h^{3/2}), \\ e^{rh} &= 1 + rh + O(h^2), \\ q_h &= \frac{2\sigma\sqrt{h} + (r - \sigma^2/2)h + O(h^{3/2})}{2\sigma\sqrt{h} + O(h^{3/2})} \\ &= \tfrac{1}{2} + \frac{r - \sigma^2/2}{2\sigma}\sqrt{h} + O(h). \end{aligned}$$

As $h \downarrow 0$ with $Nh = T$ fixed, the log-price converges in distribution toward a process with drift $r - \sigma^2/2$ and diffusion coefficient σ.

5.2 Brownian motion and stochastic calculus

Definition 5.1 (Standard Brownian motion)**.** A process $(W_t)_{t\geq 0}$ is standard Brownian motion if $W_0 = 0$ almost surely, its sample paths are continuous, it has independent increments, and $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $0 \leq s < t$.

An *Itô process* has the differential representation

$$\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t,$$

interpreted through $X_t = X_0 + \int_0^t a_s\,\mathrm{d}s + \int_0^t b_s\,\mathrm{d}W_s$. For a smooth function $f(t, x)$, Itô's formula is

$$\mathrm{d}f(t, X_t) = \Big(f_t + a_t f_x + \tfrac{1}{2}b_t^2 f_{xx}\Big)\,\mathrm{d}t + b_t f_x\,\mathrm{d}W_t.$$

The extra second-order derivative is forced by the fact that Brownian increments are of size $\sqrt{\mathrm{d}t}$, so their squares contribute at order $\mathrm{d}t$. Formally, $(\mathrm{d}W_t)^2 = \mathrm{d}t$, $\mathrm{d}W_t\,\mathrm{d}t = 0$, and $(\mathrm{d}t)^2 = 0$.

5.3 The Black–Scholes stock model

Under a physical probability measure, the geometric Brownian motion model is

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t^{\mathbb{P}}.$$

Applying Itô's formula to $f(s) = \log s$ yields

$$\begin{aligned}\mathrm{d}\log S_t &= \frac{1}{S_t}\,\mathrm{d}S_t - \frac{1}{2S_t^2}(\,\mathrm{d}S_t)^2 \\ &= \left(\mu - \frac{1}{2}\sigma^2\right)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{P}}.\end{aligned}$$

Integration and exponentiation give the exact solution

$$S_t = S_0 \exp\left\{\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t^{\mathbb{P}}\right\}.$$

Under the risk-neutral measure, the non-dividend-paying stock has $\mu = r$. The associated pricing identity for an integrable claim H is

$$V_t = \mathbb{E}^{\mathbb{Q}}\left[e^{-r(T-t)}H \mid \mathcal{F}_t\right].$$

5.4 Brownian scaling and martingale structure

Brownian motion possesses the scaling property $(W_{ct})_{t\ge 0} \overset{d}{=} (\sqrt{c}\,W_t)_{t\ge 0}$ for every $c > 0$. To verify its finite-dimensional distributions, note that both processes have zero means and covariance

$$\mathrm{Cov}(W_{cs}, W_{ct}) = c\min(s,t) = \mathrm{Cov}(\sqrt{c}\,W_s, \sqrt{c}\,W_t).$$

Since both are centered Gaussian processes with continuous paths, their laws agree. Scaling explains why increments are of order $\sqrt{h}$ rather than order h: $W_{t+h} - W_t \sim \sqrt{h}\,Z$. As $h \to 0$, the ratio $(W_{t+h} - W_t)/h$ has variance $1/h \to \infty$, indicating why classical pathwise differentiation is impossible.

Brownian motion is a martingale because for $s < t$,

$$\begin{aligned}\mathbb{E}[W_t \mid \mathcal{F}_s] &= \mathbb{E}[W_s + (W_t - W_s) \mid \mathcal{F}_s] \\ &= W_s + \mathbb{E}[W_t - W_s] = W_s.\end{aligned}$$

The process $W_t^2 - t$ is likewise a martingale, but this second fact requires knowing that the conditional second moment of a Brownian increment is $t - s$:

$$\begin{aligned}\mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] &= W_s^2 + 2W_s\mathbb{E}[W_t - W_s] + \mathbb{E}[(W_t - W_s)^2] - t \\ &= W_s^2 + (t-s) - t = W_s^2 - s.\end{aligned}$$

These are prototypes for the stochastic integral identities proved later.

5.5 Dynamics of a nonlinear stock transform

If S_t follows GBM and $Y_t = S_t^\alpha$ for a real exponent α, then $f(s) = s^\alpha$ has first two derivatives $\alpha s^{\alpha-1}$ and $\alpha(\alpha-1)s^{\alpha-2}$. Itô's formula gives

$$\begin{aligned} \mathrm{d}Y_t &= \alpha S_t^{\alpha-1}(\mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t) \\ &\quad + \tfrac{1}{2}\alpha(\alpha-1)S_t^{\alpha-2}\sigma^2 S_t^2\,\mathrm{d}t \\ &= Y_t\left[\left(\alpha\mu + \tfrac{1}{2}\alpha(\alpha-1)\sigma^2\right)\,\mathrm{d}t + \alpha\sigma\,\mathrm{d}W_t\right]. \end{aligned}$$

The variance-driven drift correction is present whenever f is nonlinear. For $\alpha = 1$ the correction disappears, but for $\alpha = 2$ it contributes $\sigma^2 S_t^2\,\mathrm{d}t$.

Lecture 6

Probability Review, Black–Scholes, and Monte Carlo

6.1 Normal and lognormal random variables

If $Z \sim \mathcal{N}(0,1)$, its density is $\varphi(z) = (2\pi)^{-1/2} e^{-z^2/2}$. The moment-generating function follows by completing the square:

$$\begin{aligned}\mathbb{E}[e^{tZ}] &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{tz - z^2/2}\, \mathrm{d}z \\ &= e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-(z-t)^2/2}\, \mathrm{d}z = e^{t^2/2}.\end{aligned}$$

Thus, for $X \sim \mathcal{N}(m, v)$, $\mathbb{E}[e^{tX}] = e^{tm + t^2 v/2}$. Consequently, if $S_T = S_0 e^{(r - \sigma^2/2)T + \sigma\sqrt{T} Z}$,

$$\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{(r-\sigma^2/2)T} \mathbb{E}[e^{\sigma\sqrt{T} Z}] = S_0 e^{rT}.$$

This is precisely the risk-neutral stock martingale requirement after discounting.

6.2 Exact simulation of Brownian motion and geometric Brownian motion

On a grid $0 = t_0 < \cdots < t_N = T$, sample independent $Z_j \sim \mathcal{N}(0,1)$. Then

$$\begin{aligned}W_{t_{j+1}} &= W_{t_j} + \sqrt{t_{j+1} - t_j}\, Z_{j+1}, \\ S_{t_{j+1}} &= S_{t_j} \exp\left\{ (r - \tfrac{1}{2}\sigma^2)(t_{j+1} - t_j) + \sigma \sqrt{t_{j+1} - t_j}\, Z_{j+1} \right\}.\end{aligned}$$

These updates are exact at the chosen observation dates in the GBM model; the randomness in between grid points is not needed for discretely sampled Asian payoffs, but it can matter for continuously monitored barriers and extrema.

Example 6.1 (Asian call Monte Carlo)**.** For m equally spaced fixing dates t_j, simulate n independent GBM paths. For each path i, compute

$$Y_i = e^{-rT} \left(\frac{1}{m} \sum_{j=1}^{m} S_{t_j}^{(i)} - K \right)^+.$$

Then $\widehat{V}_n = n^{-1} \sum_{i=1}^{n} Y_i$ estimates the risk-neutral option value. Since the averaging and positive-part operations are nonlinear, one must compute the payoff *on each path before averaging across paths.*

6.3 Statistical foundations of crude Monte Carlo

Let $Y_1, \dots, Y_n$ be independent, identically distributed, and integrable, with target $\theta = \mathbb{E}[Y_1]$. Define $\widehat{\theta}_n = n^{-1} \sum_{i=1}^n Y_i$. By linearity, $\mathbb{E}[\widehat{\theta}_n] = \theta$. If $\mathrm{Var}(Y_1) = v < \infty$, independence gives

$$\mathrm{Var}(\widehat{\theta}_n) = \frac{1}{n^2} \sum_{i=1}^n \mathrm{Var}(Y_i) = \frac{v}{n}.$$

The strong law of large numbers gives $\widehat{\theta}_n \to \theta$ almost surely. The central limit theorem gives

$$\sqrt{n} \frac{\widehat{\theta}_n - \theta}{\sqrt{v}} \Rightarrow \mathcal{N}(0,1).$$

Replacing v with $s_n^2 = (n-1)^{-1} \sum_{i=1}^n (Y_i - \widehat{\theta}_n)^2$ gives an approximate two-sided confidence interval

$$\left[\widehat{\theta}_n - z_{1-\alpha/2} \frac{s_n}{\sqrt{n}},\ \widehat{\theta}_n + z_{1-\alpha/2} \frac{s_n}{\sqrt{n}} \right].$$

To halve the standard error, the number of independent samples must be multiplied by four. This slow $O(n^{-1/2})$ rate motivates variance reduction.

6.4 Risk-neutral European option valuation: a normal integral

Under the no-dividend Black–Scholes model,

$$S_T = s \exp\left((r - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau} Z \right), \qquad Z \sim \mathcal{N}(0,1), \quad \tau = T - t.$$

Define $a = (\log(K/s) - (r - \sigma^2/2)\tau)/(\sigma\sqrt{\tau}) = -d_2$. Since $S_T > K$ exactly when $Z > a$, the call price is

$$\begin{aligned} C(t,s) &= e^{-r\tau} \mathbb{E}[(S_T - K)^+] \\ &= e^{-r\tau} \left(\mathbb{E}[S_T \mathbf{1}_{\{Z>a\}}] - K \mathbb{P}(Z > a) \right). \end{aligned}$$

The truncated lognormal expectation is

$$\begin{aligned} \mathbb{E}[S_T \mathbf{1}_{\{Z>a\}}] &= s e^{(r-\sigma^2/2)\tau} \int_a^\infty e^{\sigma\sqrt{\tau} z} \varphi(z)\, \mathrm{d}z \\ &= s e^{(r-\sigma^2/2)\tau} e^{\sigma^2\tau/2} \int_{a-\sigma\sqrt{\tau}}^\infty \varphi(y)\, \mathrm{d}y \\ &= s e^{r\tau} \Phi(\sigma\sqrt{\tau} - a) = s e^{r\tau} \Phi(d_1). \end{aligned}$$

Also, $\mathbb{P}(Z > a) = \Phi(d_2)$, so the resulting call value is $s\Phi(d_1) - K e^{-r\tau} \Phi(d_2)$. This explicit derivation explains why pricing a European call requires a *truncated* lognormal moment, not merely $\mathbb{E}[S_T]$.

6.5 Estimator design and confidence intervals for options

A Monte Carlo estimate should report more than a point value. Suppose the simulated discounted payoffs are $Y_1, \dots, Y_n$, where Y_i may be heavily right-skewed for a deep-out-of-the-money call. The

sample standard error is

$$\widehat{\mathrm{SE}}(\widehat{V}_n) = \frac{s_Y}{\sqrt{n}}, \qquad s_Y^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \widehat{V}_n)^2.$$

If the payoff has finite variance and n is sufficiently large for the normal approximation to be reliable, the interval $\widehat{V}_n \pm 1.96 s_Y/\sqrt{n}$ has approximately 95% coverage. For rare-event options, most Y_i may be zero. In this situation the CLT can converge slowly in practice and an apparently tight interval from a small sample may be misleading. A simulation diagnostic should include nonzero-payoff frequency, independent-run replicates, and stability as n increases.

Lecture 7

Random Number Generation and Monte Carlo Inversion

7.1 Pseudo-random numbers

A pseudo-random number generator is a deterministic algorithm designed to produce numbers behaving approximately like independent $U(0,1)$ random variables. For example, a linear congruential generator uses

$$X_{k+1} = (aX_k + c) \bmod m, \qquad U_k = \frac{X_k}{m}.$$

Its sequence repeats after at most m values; poor choices of (a, c, m) can lead to short periods and severe lattice artifacts. Famous defective generators, including RANDU, illustrate why marginal uniformity does not suffice: triples of successive outputs can lie on a small number of planes. Financial simulation is sensitive to the *joint* properties of the generated numbers, especially for high-dimensional path-dependent claims.

7.2 Inverse transform sampling

Theorem 7.1 (Probability integral/inverse transform)**.** *Let F be a cumulative distribution function and let $U \sim U(0,1)$. Define the generalized inverse $F^{-1}(u) = \inf\{x : F(x) \geq u\}$. Then $X = F^{-1}(U)$ has distribution function F.*

Proof. For every continuity point x of F, the generalized inverse has the event equivalence

$$\{F^{-1}(U) \leq x\} = \{U \leq F(x)\}.$$

Since U is uniform, $\mathbb{P}(X \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$. □

Example 7.2 (Exponential sampling)**.** If $X \sim \text{Exp}(\lambda)$, its CDF is $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$. Set $U = F(X)$ and solve:

$$\begin{aligned} U &= 1 - e^{-\lambda X}, \\ e^{-\lambda X} &= 1 - U, \\ X &= -\lambda^{-1} \log(1 - U). \end{aligned}$$

Because $1 - U$ is also uniform, one can equivalently simulate $X = -\log U/\lambda$, provided $U > 0$.

For distributions lacking analytically invertible CDFs, one can numerically invert F or use alternative techniques such as rejection sampling. Numerical stability in the tails matters for rare-event payoffs.

7.3 Integration as an expectation

A Lebesgue-integrable function $g : (0,1)^d \to \mathbb{R}$ defines the integral $\theta = \int_{(0,1)^d} g(u)\,\mathrm{d}u$. If $U \sim U((0,1)^d)$, then $\theta = \mathbb{E}[g(U)]$. This converts a deterministic high-dimensional integration task to a sampling problem. The Monte Carlo estimate $n^{-1}\sum_i g(U_i)$ is consistent under integrability and admits a classical CLT under finite variance. Inverse CDFs, conditional distributions, and normal transforms allow more general pricing expectations to be expressed in this form.

7.4 A formal proof of the Monte Carlo central limit scaling

Suppose Y_i are i.i.d. with mean θ and finite positive variance v. Set $Z_i = (Y_i - \theta)/\sqrt{v}$. Then $\mathbb{E}[Z_i] = 0$, $\mathrm{Var}(Z_i) = 1$ and

$$\frac{\sqrt{n}(\widehat{\theta}_n - \theta)}{\sqrt{v}} = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} Z_i.$$

The classical Lindeberg–Lévy central limit theorem yields convergence in distribution to $\mathcal{N}(0,1)$. The sample variance s_n^2 converges in probability to v under finite second moment, so Slutsky's theorem permits replacing the unknown $\sqrt{v}$ by s_n when constructing confidence intervals. This is an *asymptotic* statement and does not guarantee accurate nominal coverage for small samples or unusually heavy-tailed payoffs.

7.5 Computing expectations with one uniform variate

For a measurable g and a random variable X with CDF F, the inverse transform representation gives $X \overset{d}{=} F^{-1}(U)$ and

$$\mathbb{E}[g(X)] = \int_0^1 g(F^{-1}(u))\,\mathrm{d}u,$$

provided $g(X)$ is integrable. This formula unifies numerical quadrature and Monte Carlo. A deterministic quadrature rule selects a structured collection of u values; ordinary Monte Carlo selects independent uniforms; quasi-Monte Carlo selects a low-discrepancy set. In multidimensional simulation, one needs a mapping from uniform vectors into the joint law of the entire stochastic path, not just the correct marginals.

Example 7.3 (Exponential transform and a check)**.** Let $U \sim U(0,1)$ and $X = -\log U/\lambda$. Compute its first moment directly:

$$\begin{aligned}\mathbb{E}[X] &= -\frac{1}{\lambda}\int_0^1 \log u\,\mathrm{d}u \\ &= -\frac{1}{\lambda}\left[u\log u - u\right]_{u=0}^{u=1} = \frac{1}{\lambda}.\end{aligned}$$

A corresponding second-moment calculation gives $\mathbb{E}[X^2] = 2/\lambda^2$, hence $\mathrm{Var}(X) = 1/\lambda^2$. These moments provide useful numerical unit tests for an exponential random variate generator.

Lecture 8

Variance Reduction: Antithetic and Control Variates

8.1 Antithetic sampling

To estimate $\theta = \mathbb{E}[g(U)]$ for $U \sim U(0,1)$, sample independent $U_1, \dots, U_m$ and construct paired averages

$$A_i = \frac{g(U_i) + g(1-U_i)}{2}, \qquad \widehat{\theta}_{\mathrm{AV}} = \frac{1}{m}\sum_{i=1}^{m} A_i.$$

Because $1 - U \sim U(0,1)$, $\mathbb{E}[A_i] = \theta$. At equal computational budgets, compare this $2m$-evaluation estimator with $2m$ independent crude samples:

$$\begin{aligned}
\mathrm{Var}(\widehat{\theta}_{\mathrm{AV}}) &= \frac{1}{4m}\left[\mathrm{Var}(g(U)) + \mathrm{Var}(g(1-U)) + 2\,\mathrm{Cov}(g(U), g(1-U))\right] \\
&= \frac{v + \mathrm{Cov}(g(U), g(1-U))}{2m}, \\
\mathrm{Var}(\widehat{\theta}_{\mathrm{MC},2m}) &= \frac{v}{2m}.
\end{aligned}$$

Antithetic sampling improves precision only when the covariance of the paired payoffs is negative; this often holds for monotone g, since $g(U)$ and $g(1-U)$ then move in opposite directions. For nonmonotone g the covariance can be positive and the estimator can perform worse.

Example 8.1 (Integrating a quadratic)**.** For $g(u) = u^2$, $\theta = 1/3$ and

$$\begin{aligned}
\mathrm{Var}(U^2) &= \mathbb{E}[U^4] - (\mathbb{E}[U^2])^2 = \frac{1}{5} - \frac{1}{9} = \frac{4}{45}, \\
\mathrm{Cov}(U^2, (1-U)^2) &= \mathbb{E}[U^2(1-U)^2] - \frac{1}{9} \\
&= \left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5}\right) - \frac{1}{9} = \frac{1}{30} - \frac{1}{9} = -\frac{7}{90}.
\end{aligned}$$

Consequently, $\mathrm{Var}(A_i) = \frac{1}{2}(4/45 - 7/90) = 1/180$, a substantial reduction from the variance $4/45$ of a single evaluation.

8.2 Control variates

Suppose Y is the target payoff and C is a correlated random variable with a *known* expectation $c = \mathbb{E}[C]$. For any fixed coefficient $b \in \mathbb{R}$, define

$$Z_b = Y - b(C - c), \qquad \mathbb{E}[Z_b] = \mathbb{E}[Y] = \theta.$$

Its variance is

$$\operatorname{Var}(Z_b) = \operatorname{Var}(Y) + b^2 \operatorname{Var}(C) - 2b \operatorname{Cov}(Y, C).$$

Differentiating this quadratic in b gives

$$b^* = \frac{\operatorname{Cov}(Y, C)}{\operatorname{Var}(C)}, \qquad \operatorname{Var}(Z_{b^*}) = \operatorname{Var}(Y)\left(1 - \operatorname{Corr}(Y, C)^2\right),$$

when $\operatorname{Var}(C) > 0$. Thus any nonzero correlation provides theoretical variance reduction with the optimal coefficient. In practice b^* is unknown; it can be estimated from an independent pilot sample to preserve exact unbiasedness of the main estimator conditional on that pilot.

Example 8.2 (Asian option control)**.** For an arithmetic-average Asian payoff Y, a geometric-average Asian payoff can be a useful control because it is driven by the same path and often has a known analytic model price. Draw Y_i and C_i using identical stock paths, estimate b^*, and form

$$\widehat{V}_{\mathrm{CV}} = \overline{Y} - b(\overline{C} - c).$$

The expected discounted geometric payoff c must correspond exactly to the same observation schedule, model, and discounting conventions.

8.3 Several controls and practical confidence intervals

For a control vector $C \in \mathbb{R}^k$ with known mean c, minimize $\operatorname{Var}(Y - b^\top(C - c))$ over b. Under invertibility of $\Sigma_{CC} = \operatorname{Cov}(C, C)$,

$$b^* = \Sigma_{CC}^{-1} \operatorname{Cov}(C, Y).$$

A multivariate regression of Y on C estimates these coefficients. When calculating error bars for an antithetic estimator, treat the *pair means* A_i as independent sampling units; treating all $2m$ evaluations as independent ignores within-pair correlation.

8.4 Why antithetic covariances can have either sign

The efficacy of antithetic sampling depends on the shape of g. If g is increasing, then $g(U)$ increases with U while $g(1 - U)$ decreases. A general rearrangement/covariance inequality then implies $\operatorname{Cov}(g(U), g(1-U)) \le 0$ when the required moments exist. If g is nonmonotone, the two transforms can move together. For example, set $g(u) = (u - 1/2)^2$. Then $g(1 - u) = g(u)$ for every u, so

$$\operatorname{Cov}(g(U), g(1 - U)) = \operatorname{Var}(g(U)) > 0.$$

At equal total payoff evaluations, pairing identical evaluations uses half as many independent inputs and doubles the estimator variance relative to independent Monte Carlo. This counterexample shows that the word *antithetic* does not itself guarantee variance reduction.

8.5 Optimal controls derived from a projection

Let $\widetilde{Y} = Y - \mathbb{E}[Y]$ and $\widetilde{C} = C - \mathbb{E}[C]$ be centered square-integrable variables. Think of L^2 as an inner-product space with $\langle X, Z\rangle = \mathbb{E}[XZ]$. The optimal scalar control coefficient minimizes

$$\left\|\widetilde{Y} - b\widetilde{C}\right\|_{L^2}^2.$$

Orthogonality of the residual to the control demands

$$\begin{aligned} 0 &= \langle \widetilde{Y} - b^*\widetilde{C}, \widetilde{C}\rangle \\ &= \mathrm{Cov}(Y, C) - b^* \,\mathrm{Var}(C), \\ b^* &= \frac{\mathrm{Cov}(Y, C)}{\mathrm{Var}(C)}. \end{aligned}$$

This is the population least-squares regression coefficient. For k controls, orthogonality to every centered control produces the normal equations $\Sigma_{CC} b = \mathrm{Cov}(C, Y)$. Strong correlation means that the payoff's random fluctuations can be predicted from control fluctuations with known expectation and subsequently subtracted.

8.6 Cost-aware variance reduction

Suppose method A has per-sample cost c_A and variance v_A for an independent sample unit, whereas method B has cost c_B and variance v_B. At a fixed total computational budget B, their approximate squared errors are proportional to $v_A c_A/B$ and $v_B c_B/B$. Consequently, the appropriate efficiency comparison is

$$\text{Efficiency ratio} = \frac{v_A c_A}{v_B c_B},$$

not merely the two variances at unequal computational effort. Antithetic sampling can yield a dramatic variance reduction while using paired evaluations; controls may require substantial pilot-estimation or analytic computation. Performance claims should always include cost as well as precision.

Lecture 10

Correlated Simulation, Dynamic Hedging, and Risk Measures

10.1 Multivariate Gaussian simulation

Let $Z = (Z_1, \dots, Z_d)^\top$ have independent $\mathcal{N}(0,1)$ coordinates. If $\mu \in \mathbb{R}^d$ and $A \in \mathbb{R}^{d\times d}$, then

$$X = \mu + AZ, \qquad \mathbb{E}[X] = \mu, \qquad \mathrm{Cov}(X) = AA^\top.$$

Given a positive-definite covariance matrix Σ, the Cholesky decomposition supplies a lower-triangular L such that $\Sigma = LL^\top$. Hence $X = \mu + LZ \sim \mathcal{N}_d(\mu, \Sigma)$. If Σ is merely positive semidefinite, an eigenvalue or singular-value factorization provides a suitable rectangular square root.

Example 10.1 (Two correlated standard normals)**.** For target correlation $\rho \in [-1, 1]$, take independent $Z_1, Z_2 \sim \mathcal{N}(0,1)$ and set

$$X_1 = Z_1, \qquad X_2 = \rho Z_1 + \sqrt{1-\rho^2}\, Z_2.$$

Then $\mathrm{Var}(X_1) = \mathrm{Var}(X_2) = 1$ and

$$\begin{aligned}\mathrm{Cov}(X_1, X_2) &= \mathbb{E}\left[Z_1(\rho Z_1 + \sqrt{1-\rho^2} Z_2)\right] \\ &= \rho\mathbb{E}[Z_1^2] + \sqrt{1-\rho^2}\mathbb{E}[Z_1 Z_2] = \rho.\end{aligned}$$

For a two-asset risk-neutral diffusion driven by correlated Brownian motions,

$$\frac{\mathrm{d}S_t^{(i)}}{S_t^{(i)}} = r\,\mathrm{d}t + \sigma_i\,\mathrm{d}W_t^{(i)}, \qquad \mathrm{d}[W^{(1)}, W^{(2)}]_t = \rho\,\mathrm{d}t,$$

apply the normal construction independently at each time step; independent draws across steps preserve the Brownian independent-increment property.

10.2 Dynamic trading and discrete delta hedging

A European derivative price $V(t,S)$ can be hedged by holding $\Delta_t = V_S(t, S_t)$ shares. If a trader sells the option, the hedge portfolio (stock plus cash minus option) is

$$\Pi_t = -V(t, S_t) + \Delta_t S_t + B_t.$$

With initial cash $B_0 = V(0, S_0) - \Delta_0 S_0$, the initial net value is zero. At rebalancing times, choose $\Delta_{t_{j+1}}$ using current information and move cash to keep the strategy self-financing. Under a discretized strategy, residual profit and loss is generally nonzero because the hedge is not continuously adjusted.

Proposition 10.2 (Leading-order unhedged option change)**.** *Over a short interval with stock change ΔS, a twice-differentiable option value satisfies*

$$\Delta V \approx V_t \Delta t + \Delta \Delta S + \tfrac{1}{2}\Gamma(\Delta S)^2, \qquad \Gamma = V_{SS}.$$

For a delta hedge fixed over the interval, the stochastic linear term is canceled, but the realized quadratic move contributes $-\frac{1}{2}\Gamma(\Delta S)^2$ to a short-option hedged portfolio, before financing and time-decay effects.

If the hedge model uses volatility $\widehat{\sigma}$ while realized instantaneous quadratic variation follows $\sigma_t^2 S_t^2\,\mathrm{d}t$, the continuous delta-hedged profit of a short European option is, under suitable assumptions,

$$\mathrm{d}\Pi_t = \tfrac{1}{2}\Gamma_t S_t^2(\widehat{\sigma}^2 - \sigma_t^2)\,\mathrm{d}t.$$

The sign reverses for a long-option position. This identity explains why realized variance relative to the model's implied variance influences the performance of delta hedging.

10.3 Gamma-neutral portfolios

Suppose, in addition to stock, one can trade another derivative $I(t, S)$ with nonzero I_{SS}. A portfolio short one unit of V, long δ_S shares, and long δ_I units of I has local delta and gamma

$$\begin{aligned}\Pi_S &= -V_S + \delta_S + \delta_I I_S,\\ \Pi_{SS} &= -V_{SS} + \delta_I I_{SS}.\end{aligned}$$

Imposing both equal to zero yields

$$\delta_I = \frac{V_{SS}}{I_{SS}}, \qquad \delta_S = V_S - \frac{V_{SS}}{I_{SS}} I_S.$$

The remaining cash holding makes the overall strategy self-financing. This hedge is local and depends on both derivatives' sensitivities, which can change over time.

10.4 Value-at-Risk and Expected Shortfall

Let L represent monetary *loss* over a fixed horizon, so that large positive values are undesirable. For a confidence level $\alpha \in (0, 1)$, define

$$\mathrm{VaR}_\alpha(L) = \inf\{\ell \in \mathbb{R} : \mathbb{P}(L \le \ell) \ge \alpha\}.$$

For an integrable loss with a continuous distribution at the α-quantile,

$$\mathrm{ES}_\alpha(L) = \mathbb{E}[L \mid L \ge \mathrm{VaR}_\alpha(L)].$$

A more general tail-average definition, suitable in the presence of atoms, is

$$\mathrm{ES}_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 \mathrm{VaR}_u(L)\,\mathrm{d}u.$$

For $L \sim \mathcal{N}(m, s^2)$, put $z_\alpha = \Phi^{-1}(\alpha)$ and note $L = m + sZ$. Then

$$\mathrm{VaR}_\alpha(L) = m + s z_\alpha,$$
$$\mathrm{ES}_\alpha(L) = m + s\frac{\varphi(z_\alpha)}{1-\alpha}.$$

Indeed, $\mathbb{E}[Z\mathbf{1}_{\{Z>z\}}] = \int_z^\infty x\varphi(x)\,\mathrm{d}x = \varphi(z)$. For simulated losses $L_1, \dots, L_n$, estimate VaR by the empirical quantile and ES by the average of the worst tail observations (with appropriate treatment of the cutoff). Tail estimators can have considerable sampling uncertainty, particularly for large α.

10.5 Portfolio correlation and pathwise dependence

Correlation must be imposed at the level of the driving shocks, not by merely correlating terminal outputs after the simulation. Suppose $\mathrm{d}W_1\,\mathrm{d}W_2 = \rho\,\mathrm{d}t$ in the Itô multiplication notation, and let two stocks solve

$$\mathrm{d}S_t^{(1)} = \mu_1 S_t^{(1)}\,\mathrm{d}t + \sigma_1 S_t^{(1)}\,\mathrm{d}W_t^{(1)}, \qquad \mathrm{d}S_t^{(2)} = \mu_2 S_t^{(2)}\,\mathrm{d}t + \sigma_2 S_t^{(2)}\,\mathrm{d}W_t^{(2)}.$$

For log prices, $\mathrm{d}\log S_t^{(i)} = (\mu_i - \sigma_i^2/2)\,\mathrm{d}t + \sigma_i\,\mathrm{d}W_t^{(i)}$. Therefore

$$\begin{aligned}\mathrm{Cov}\left(\log\frac{S_T^{(1)}}{S_0^{(1)}}, \log\frac{S_T^{(2)}}{S_0^{(2)}}\right) &= \sigma_1\sigma_2\,\mathrm{Cov}(W_T^{(1)}, W_T^{(2)}) \\ &= \rho\sigma_1\sigma_2 T.\end{aligned}$$

Yet the *correlation of the stock returns* is not exactly ρ for finite T: exponentiation changes correlation. This matters when constructing basket option payoffs and estimating tail dependence. The driver correlation ρ belongs to the joint normal Brownian increments, while nonlinear transformations generate a different dependence measure for the resulting asset prices.

10.6 Risk estimation with a simulated loss sample

To estimate the loss distribution of a hedged option book, simulate market paths under an appropriate scenario law, update the derivatives and hedge positions at the required trading dates, and record the terminal change in value net of cash financing and any specified transaction costs. The distribution used for *risk* calculations is typically a physical or stress-scenario distribution, not automatically the same as the risk-neutral measure used for pricing. A properly implemented Monte Carlo workflow must distinguish the probability law under which it computes expected discounted payoffs from the law under which it estimates realized losses.

For example, if L_i are i.i.d. simulated portfolio losses and $L_{(1)} \le \cdots \le L_{(n)}$ are their order statistics, an empirical α-VaR estimate is $L_{(\lceil n\alpha \rceil)}$. If the upper tail has m observations and the distribution is sufficiently smooth, a simple ES estimator averages the m largest losses. In the presence of ties or a desired exact tail fraction $(1-\alpha)$, the cutoff observation may receive fractional weight. Both estimators depend strongly on the quality of the simulated tail, which is particularly challenging for portfolios short convex options.

10.7 Numerical Greeks by bump-and-revalue

If a derivative is priced by a numerical routine $V(s, \sigma, r, t)$, finite differences offer approximate sensitivities without requiring a closed formula. Holding all other inputs fixed, a central-difference delta is

$$\widehat{\Delta}_h = \frac{V(s+h, \sigma, r, t) - V(s-h, \sigma, r, t)}{2h},$$

and a central-difference gamma is

$$\widehat{\Gamma}_h = \frac{V(s+h, \sigma, r, t) - 2V(s, \sigma, r, t) + V(s-h, \sigma, r, t)}{h^2}.$$

Taylor expansion shows that if V is sufficiently smooth and the prices are exact, both formulas have truncation error $O(h^2)$. In a Monte Carlo pricer, however, each evaluated price has sampling noise. A small h reduces Taylor bias but amplifies uncorrelated simulation noise by factors of order h^{-1} for delta and h^{-2} for gamma. It is therefore often advantageous to use the same random input paths for the bumped prices, inducing positive covariance that partially cancels when differences are formed.

An alternative is the pathwise derivative method. Under an exact GBM simulation $S_T = sF(Z)$ with $F(Z) > 0$ independent of s, a smooth European payoff H permits, under suitable differentiation-under-the-expectation assumptions,

$$\begin{aligned}\Delta &= \frac{\partial}{\partial s}\left(e^{-rT}\mathbb{E}[H(sF(Z))]\right) \\ &= e^{-rT}\mathbb{E}[H'(sF(Z))F(Z)].\end{aligned}$$

For a call, $H'(x) = \mathbf{1}_{\{x>K\}}$ away from its strike, so a valid pathwise-delta estimator is

$$\widehat{\Delta}_{\mathrm{PW}} = \frac{e^{-rT}}{n}\sum_{i=1}^{n} \mathbf{1}_{\{S_T^{(i)}>K\}}\frac{S_T^{(i)}}{s}.$$

The kink does not prevent this first derivative under the continuous lognormal model, because the event $S_T = K$ has probability zero. The analogous pathwise second derivative is more delicate because the payoff derivative is discontinuous; blindly differentiating the indicator gives an incorrect zero gamma.

Lecture 11

Quadratic Variation and Construction of the Itô Integral

11.1 From ordinary to stochastic integration

For a continuously differentiable path $x : [0, T] \to \mathbb{R}$, ordinary calculus relates its increments to $\int_0^T x'(t)\,\mathrm{d}t$. Brownian paths, however, are almost surely nowhere differentiable, so $\int H_t\,\mathrm{d}W_t$ cannot be interpreted through the classical Riemann–Stieltjes integral for arbitrary adapted integrands. The correct construction begins with simple predictable processes and then passes to a suitable L^2 limit.

11.2 Quadratic variation of Brownian motion

Let $\pi_n = \{0 = t_0^{(n)} < \cdots < t_{m_n}^{(n)} = t\}$ be a sequence of partitions whose maximum mesh tends to zero. Its discrete quadratic variation is

$$Q_{\pi_n}(W) = \sum_{j=0}^{m_n-1} (W_{t_{j+1}^{(n)}} - W_{t_j^{(n)}})^2.$$

Theorem 11.1. *For Brownian motion, $Q_{\pi_n}(W) \to t$ in L^2 whenever the partition mesh tends to zero. In particular $[W]_t = t$.*

Proof. Write $\Delta_j W = W_{t_{j+1}} - W_{t_j}$ and $h_j = t_{j+1} - t_j$. Independence and $\Delta_j W \sim \mathcal{N}(0, h_j)$ give

$$\begin{aligned}
\mathbb{E}[Q_{\pi_n}(W)] &= \sum_j \mathbb{E}[(\Delta_j W)^2] = \sum_j h_j = t, \\
\operatorname{Var}(Q_{\pi_n}(W)) &= \sum_j \operatorname{Var}((\Delta_j W)^2) \\
&= \sum_j (\mathbb{E}[(\Delta_j W)^4] - h_j^2) \\
&= 2\sum_j h_j^2 \le 2\,|\pi_n| \sum_j h_j = 2t\,|\pi_n| \longrightarrow 0.
\end{aligned}$$

Thus $\mathbb{E}[(Q_{\pi_n}(W) - t)^2] \to 0$. □

A continuously differentiable path f has zero quadratic variation because

$$\sum_j (f(t_{j+1}) - f(t_j))^2 \le \Big(\max_j |f(t_{j+1}) - f(t_j)|\Big) \sum_j |f(t_{j+1}) - f(t_j)| \to 0.$$

Consequently, time has $[t]_T = 0$ but Brownian motion has positive quadratic variation. This distinction is what changes the chain rule.

11.3 Simple predictable integrands

Definition 11.2 (Itô integral of a simple process)**.** Let $H_t = \sum_{j=0}^{n-1} H_j \mathbf{1}_{(t_j, t_{j+1}]}(t)$ with H_j measurable with respect to $\mathcal{F}_{t_j}$ and $\mathbb{E}[H_j^2] < \infty$. Define

$$\int_0^T H_t \, \mathrm{d}W_t = \sum_{j=0}^{n-1} H_j (W_{t_{j+1}} - W_{t_j}).$$

The use of *left-endpoint* information matters: H_j must not depend on the future increment it multiplies. Conditioned on $\mathcal{F}_{t_j}$, the next Brownian increment has mean zero, so the integral has zero expectation.

Theorem 11.3 (Itô isometry)**.** *For every square-integrable predictable integrand H,*

$$\mathbb{E}\left[\left(\int_0^T H_t \, \mathrm{d}W_t\right)^2\right] = \mathbb{E}\left[\int_0^T H_t^2 \, \mathrm{d}t\right].$$

Proof. For simple H, expand the square of the sum. Cross terms vanish because earlier terms are measurable at the start of a later increment, which has conditional mean zero. The diagonal terms satisfy

$$\begin{aligned}\mathbb{E}\left[H_j^2 (\Delta_j W)^2\right] &= \mathbb{E}\left[H_j^2 \mathbb{E}[(\Delta_j W)^2 \mid \mathcal{F}_{t_j}]\right] \\ &= h_j \mathbb{E}[H_j^2].\end{aligned}$$

Summing proves the identity for simple processes. Approximate a general predictable H in $L^2(\Omega \times [0,T])$ by simple predictable processes; the isometry makes their stochastic integrals Cauchy in $L^2(\Omega)$ and defines the limiting integral. □

11.4 Riemann and Lebesgue–Stieltjes integration

For a smooth deterministic integrator F with derivative F', classical Riemann–Stieltjes integration reduces to

$$\int_a^b f(x) \, \mathrm{d}F(x) = \int_a^b f(x) F'(x) \, \mathrm{d}x.$$

When F is a cumulative distribution function, the Lebesgue–Stieltjes integral $\int g(x) \, \mathrm{d}F(x)$ is the expectation $\mathbb{E}[g(X)]$ if X has CDF F and $g(X)$ is integrable. This integral encompasses both continuous and discrete distributions. If X takes values x_j with probabilities p_j, then

$$\int_{\mathbb{R}} g(x) \, \mathrm{d}F(x) = \sum_j g(x_j) p_j.$$

If X has density f, the same expression becomes $\int_{\mathbb{R}} g(x) f(x) \, \mathrm{d}x$. In either case the CDF is a probability measure's cumulative mass, so expectations can be written without choosing a density or probability-mass representation.

Brownian sample paths are continuous but have infinite total variation almost surely on every nontrivial interval. Therefore, trying to treat W_t as a bounded-variation Stieltjes integrator fails. The Itô integral is defined probabilistically, with left-endpoint adapted integrands and an L^2 completion, rather than by naive pathwise Riemann–Stieltjes limits.

11.5 A Brownian quadratic-variation calculation with an explicit error bound

For an equidistant grid $t_j = jT/n$,

$$Q_n = \sum_{j=1}^{n} (W_{t_j} - W_{t_{j-1}})^2.$$

Each increment has variance T/n, so $\mathbb{E}[Q_n] = T$. Because a centered normal $Z \sim \mathcal{N}(0, v)$ satisfies $\mathbb{E}[Z^4] = 3v^2$,

$$\mathrm{Var}(Q_n) = n\left(3\frac{T^2}{n^2} - \frac{T^2}{n^2}\right) = \frac{2T^2}{n},$$
$$\mathbb{E}[(Q_n - T)^2] = \frac{2T^2}{n}.$$

By Chebyshev's inequality,

$$\mathbb{P}(|Q_n - T| > \varepsilon) \le \frac{2T^2}{n\varepsilon^2}.$$

This gives a quantitative probability bound for discretized quadratic variation. The result is useful both as a proof tool and as a practical simulation diagnostic: aggregating squared Brownian increments over increasingly fine partitions should approach elapsed time.

Lecture 12

Itô's Formula and Stochastic Integral Examples

12.1 Why the classical chain rule fails

Take a smooth function f and expand over a short Brownian increment:

$$f(W_{t+h}) - f(W_t) = f'(W_t)\Delta W_t + \tfrac{1}{2}f''(W_t)(\Delta W_t)^2 + R_h.$$

Summing across a fine partition, the first term converges to the Itô integral and the second to $\frac{1}{2}\int_0^T f''(W_t)\,\mathrm{d}t$ because Brownian quadratic variation accumulates. The higher-order remainder vanishes under suitable regularity. Therefore

$$f(W_T) - f(W_0) = \int_0^T f'(W_t)\,\mathrm{d}W_t + \tfrac{1}{2}\int_0^T f''(W_t)\,\mathrm{d}t.$$

Theorem 12.1 (One-dimensional Itô–Doeblin formula)**.** *Suppose* $\mathrm{d}X_t = a(t, X_t)\,\mathrm{d}t + b(t, X_t)\,\mathrm{d}W_t$, *and* $f \in C^{1,2}$. *Under usual local integrability assumptions,*

$$\begin{aligned}\mathrm{d}f(t, X_t) = &\Big[f_t(t, X_t) + a(t, X_t)f_x(t, X_t) + \tfrac{1}{2}b(t, X_t)^2 f_{xx}(t, X_t)\Big]\,\mathrm{d}t \\ &+ b(t, X_t)f_x(t, X_t)\,\mathrm{d}W_t.\end{aligned}$$

Example 12.2 (A stochastic integral evaluated explicitly)**.** Apply the formula to $f(x) = x^2$. Since $f' = 2x$ and $f'' = 2$,

$$\begin{aligned}\mathrm{d}W_t^2 &= 2W_t\,\mathrm{d}W_t + \mathrm{d}t, \\ W_T^2 - W_0^2 &= 2\int_0^T W_t\,\mathrm{d}W_t + T, \\ \int_0^T W_t\,\mathrm{d}W_t &= \frac{W_T^2 - T}{2}.\end{aligned}$$

For $f(x) = \cos x$, the same argument gives

$$\int_0^T \sin(W_t)\,\mathrm{d}W_t = 1 - \cos(W_T) - \frac{1}{2}\int_0^T \cos(W_t)\,\mathrm{d}t.$$

The second result is useful for the supplied practice problem: the stochastic integral is replaced by an ordinary time integral.

12.2 Martingales generated by stochastic integrals

If H is square-integrable and predictable, $M_t = \int_0^t H_s\,\mathrm{d}W_s$ is a martingale with respect to the Brownian filtration. In particular, for $0 \le s \le t$,

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s, \qquad \mathbb{E}[M_t] = 0.$$

The representation $W_t^2 - t = 2\int_0^t W_s \, \mathrm{d}W_s$ immediately proves that $W_t^2 - t$ is a martingale, whereas W_t^2 is not. Itô's formula and quadratic variation thus explain why stochastic calculus differs from classical calculus even for simple polynomial functions.

12.3 Product formula and covariance variation

For continuous Itô processes X and Y,

$$\mathrm{d}(X_t Y_t) = X_t \, \mathrm{d}Y_t + Y_t \, \mathrm{d}X_t + \mathrm{d}[X, Y]_t.$$

If $\mathrm{d}X_t = a_X \, \mathrm{d}t + b_X \, \mathrm{d}W_t$ and $\mathrm{d}Y_t = a_Y \, \mathrm{d}t + b_Y \, \mathrm{d}W_t$, then $\mathrm{d}[X, Y]_t = b_X b_Y \, \mathrm{d}t$. Driven by independent Brownian motions, the instantaneous cross variation is zero. The product rule is the two-variable case of multidimensional Itô's formula.

12.4 The stochastic integration-by-parts identity

For sufficiently regular deterministic functions f and g, integration by parts states that $\mathrm{d}(fg) = f \, \mathrm{d}g + g \, \mathrm{d}f$. In stochastic calculus an additional term is needed. Let $X_t = W_t$ and $Y_t = W_t$, then

$$\begin{aligned} \mathrm{d}(W_t^2) &= 2W_t \, \mathrm{d}W_t + \mathrm{d}[W]_t \\ &= 2W_t \, \mathrm{d}W_t + \mathrm{d}t. \end{aligned}$$

Omitting $\mathrm{d}t$ would imply $\mathbb{E}[W_t^2] = 0$, contradicting the normal variance t. This provides a direct consistency check on the stochastic product rule and underlines why the quadratic variation term is not optional.

For deterministic, continuously differentiable h, the product $h(t)W_t$ satisfies

$$\mathrm{d}(h(t)W_t) = h'(t)W_t \, \mathrm{d}t + h(t) \, \mathrm{d}W_t,$$

with no cross-variation correction because h has zero quadratic variation. Therefore

$$\int_0^T h(t) \, \mathrm{d}W_t = h(T)W_T - h(0)W_0 - \int_0^T h'(t)W_t \, \mathrm{d}t.$$

The right-hand side involves an ordinary time integral. This representation is sometimes useful for both theoretical calculations and numerical checks of deterministic-integrand stochastic integrals.

12.5 Conditional moments of Itô integrals

If h is deterministic and square-integrable, then $I = \int_0^T h(t) \, \mathrm{d}W_t$ is Gaussian with mean zero and variance $\int_0^T h(t)^2 \, \mathrm{d}t$. One proof approximates h by deterministic step functions. Each approximating integral is a linear combination of independent normal increments, hence is normal. The Itô isometry gives convergence of means and second moments; Gaussian characteristic functions identify the limit. For example,

$$\int_0^T e^{-\kappa(T-s)} \, \mathrm{d}W_s \sim \mathcal{N}\left(0, \frac{1 - e^{-2\kappa T}}{2\kappa}\right).$$

This simple normal-integral identity provides the exact innovation distribution in the Vasicek model and is a foundational tool for Gaussian affine processes.

Lecture 13

SDE Solutions and Basic Simulation Schemes

13.1 Stochastic differential equations as integral equations

Consider

$$\mathrm{d}X_t = a(t, X_t)\,\mathrm{d}t + b(t, X_t)\,\mathrm{d}W_t, \qquad X_0 = x_0.$$

A strong solution is an adapted process satisfying

$$X_t = x_0 + \int_0^t a(s, X_s)\,\mathrm{d}s + \int_0^t b(s, X_s)\,\mathrm{d}W_s$$

almost surely. Conditions such as global Lipschitz continuity and linear growth of a and b ensure existence and pathwise uniqueness in standard settings. Numerical discretizations approximate the integral equation, not an ordinary differentiable trajectory.

13.2 Exact solutions for benchmark processes

Example 13.1 (Generalized GBM)**.** For $\mathrm{d}X_t = \alpha X_t\,\mathrm{d}t + \beta X_t\,\mathrm{d}W_t$, Itô's formula applied to $\log X_t$ gives

$$\begin{aligned}\mathrm{d}\log X_t &= \alpha\,\mathrm{d}t + \beta\,\mathrm{d}W_t - \tfrac{1}{2}\beta^2\,\mathrm{d}t,\\ X_t &= X_0 \exp\left\{(\alpha - \beta^2/2)t + \beta W_t\right\}.\end{aligned}$$

Example 13.2 (Vasicek short-rate model)**.** Suppose $\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \eta\,\mathrm{d}W_t$, where $\kappa > 0$. Multiply by $e^{\kappa t}$ and use the product rule:

$$\begin{aligned}\mathrm{d}(e^{\kappa t} r_t) &= \kappa e^{\kappa t} r_t\,\mathrm{d}t + e^{\kappa t}\,\mathrm{d}r_t\\ &= \kappa\theta e^{\kappa t}\,\mathrm{d}t + \eta e^{\kappa t}\,\mathrm{d}W_t.\end{aligned}$$

Therefore

$$r_t = \theta + (r_0 - \theta)e^{-\kappa t} + \eta \int_0^t e^{-\kappa(t-s)}\,\mathrm{d}W_s.$$

The stochastic integral is centered Gaussian with variance

$$\eta^2 \int_0^t e^{-2\kappa(t-s)}\,\mathrm{d}s = \frac{\eta^2}{2\kappa}(1 - e^{-2\kappa t}).$$

Remark 13.3 (CIR rate and variance process). The Cox–Ingersoll–Ross model takes $\mathrm{d}X_t = \kappa(\theta - X_t)\,\mathrm{d}t + \eta\sqrt{X_t}\,\mathrm{d}W_t$. Its square-root diffusion and suitable parameter conditions support nonnegative trajectories; the strict Feller condition $2\kappa\theta \geq \eta^2$ prevents an attainable zero boundary when initialized positively. A naive Gaussian Euler step can become negative even though the exact continuous-time process is nonnegative.

13.3 Euler–Maruyama discretization

Let $t_n = nh$ and $\Delta W_n = \sqrt{h} Z_n$ with independent $Z_n \sim \mathcal{N}(0,1)$. Approximating the drift and volatility as constant over each step produces

$$\widehat{X}_{n+1} = \widehat{X}_n + a(t_n, \widehat{X}_n)h + b(t_n, \widehat{X}_n)\Delta W_n.$$

This is the Euler–Maruyama scheme. Its one-step innovations depend only on present information. For GBM, Euler yields $\widehat{S}_{n+1} = \widehat{S}_n(1 + rh + \sigma\sqrt{h}Z_n)$, which can become negative, unlike the exact exponential update. When an exact transition is known, it should generally be preferred to Euler for that component.

13.4 From Itô's formula to an integrating factor

An SDE may look nonlinear in its current coordinate yet become tractable after a suitable transformation. A linear mean-reverting diffusion $\mathrm{d}X_t = \kappa(\theta - X_t)\,\mathrm{d}t + \eta\,\mathrm{d}W_t$ illustrates the general method. Rearrange as $\mathrm{d}X_t + \kappa X_t\,\mathrm{d}t = \kappa\theta\,\mathrm{d}t + \eta\,\mathrm{d}W_t$ and multiply by $e^{\kappa t}$. Since the deterministic factor has zero quadratic variation, the Itô product rule gives

$$\begin{aligned}\mathrm{d}(e^{\kappa t}X_t) &= e^{\kappa t}\,\mathrm{d}X_t + \kappa e^{\kappa t}X_t\,\mathrm{d}t \\ &= \kappa\theta e^{\kappa t}\,\mathrm{d}t + \eta e^{\kappa t}\,\mathrm{d}W_t.\end{aligned}$$

After integration, the exact conditional distribution over a step h is

$$X_{t+h} \mid X_t = x \sim \mathcal{N}\left(\theta + (x-\theta)e^{-\kappa h},\ \frac{\eta^2}{2\kappa}(1 - e^{-2\kappa h})\right).$$

This is a better simulator for Vasicek than Euler, because it avoids discretization bias at the grid points and reproduces the exact conditional moments.

13.5 Euler bias for GBM and the lognormal benchmark

Euler–Maruyama applied to $\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$ gives

$$\widehat{S}_{n+1} = \widehat{S}_n(1 + rh + \sigma\sqrt{h}Z_{n+1}).$$

Its conditional first moment is exact over one step to first order:

$$\mathbb{E}[\widehat{S}_{n+1} \mid \widehat{S}_n = s] = s(1 + rh),$$

whereas exact GBM has conditional mean $se^{rh} = s(1 + rh + r^2h^2/2 + \cdots)$. Thus the per-step weak error in the mean is $O(h^2)$ and the accumulated error over T/h steps is $O(h)$. Furthermore, the Euler multiplier may be negative when

$$Z_{n+1} < -\frac{1 + rh}{\sigma\sqrt{h}},$$

which has positive Gaussian probability for every fixed $h > 0$. A method that exactly preserves nonnegativity may therefore be preferable for pricing payoffs sensitive to extreme paths.

Lecture 14

Milstein, Higher-Order Weak Schemes, and Multivariate Itô Calculus

14.1 Itô–Taylor operators

For a time-homogeneous scalar SDE $\mathrm{d}X = a(X)\,\mathrm{d}t + b(X)\,\mathrm{d}W$, define the differential operators

$$L^0 f(x) = a(x)f'(x) + \tfrac{1}{2}b(x)^2 f''(x), \qquad L^1 f(x) = b(x)f'(x).$$

Then Itô's formula is concisely written $\mathrm{d}f(X_t) = (L^0 f)(X_t)\,\mathrm{d}t + (L^1 f)(X_t)\,\mathrm{d}W_t$. Applying this identity to the drift and diffusion coefficients inside the integral solution supplies successive correction terms.

14.2 Derivation of the Milstein update

Expand the diffusion coefficient over one short step:

$$b(X_s) \approx b(X_{t_n}) + b'(X_{t_n})b(X_{t_n})(W_s - W_{t_n}) + exthigher - orderterms.$$

Substituting into the stochastic integral gives

$$\begin{aligned}\int_{t_n}^{t_{n+1}} b(X_s)\,\mathrm{d}W_s &\approx b(X_{t_n})\Delta W_n + b'(X_{t_n})b(X_{t_n})\int_{t_n}^{t_{n+1}} (W_s - W_{t_n})\,\mathrm{d}W_s \\ &= b(X_{t_n})\Delta W_n + \tfrac{1}{2}b(X_{t_n})b'(X_{t_n})\left((\Delta W_n)^2 - h\right).\end{aligned}$$

Consequently, the Milstein approximation is

$$\widehat{X}_{n+1} = \widehat{X}_n + a_n h + b_n \Delta W_n + \frac{1}{2} b_n b_n' \left((\Delta W_n)^2 - h\right).$$

Under standard regularity assumptions, scalar Euler has strong order $1/2$ and Milstein strong order 1, reflecting the improved approximation of individual sample paths.

14.3 Strong versus weak error

Definition 14.1 (Convergence orders)**.** A scheme $\widehat{X}_T^{(h)}$ has strong order γ if $\mathbb{E}\left|X_T - \widehat{X}_T^{(h)}\right| = O(h^\gamma)$ (or the corresponding specified L^p norm), and weak order β if

$$\left|\mathbb{E}[f(X_T)] - \mathbb{E}[f(\widehat{X}_T^{(h)})]\right| = O(h^\beta)$$

for a suitable class of sufficiently smooth test functions f.

Strong convergence is important when approximating realized paths or path-dependent functionals; weak convergence is often sufficient for expected payoffs. Classical Euler–Maruyama generally has weak order 1 under smoothness conditions. A numerical method may have higher weak than strong order; the relevant criterion depends on the computational task and the payoff's regularity.

14.4 A simplified weak second-order expansion

For a scalar time-homogeneous SDE with sufficiently smooth coefficients, one useful simplified weak order-two scheme has increment

$$\begin{aligned}\widehat{X}_{n+1} - \widehat{X}_n = a_n h + b_n \Delta W_n + \tfrac{1}{2} b_n b_n' ((\Delta W_n)^2 - h) \\ + \tfrac{1}{2}(a_n' b_n + a_n b_n' + \tfrac{1}{2} b_n^2 b_n'') \Delta W_n h \\ + \tfrac{1}{2}(a_n a_n' + \tfrac{1}{2} b_n^2 a_n'') h^2.\end{aligned}$$

This is an approximation to the relevant Itô–Taylor expansion and should not be asserted to have strong order two; replacing certain multiple stochastic integrals by simplified expressions targets weak expectation accuracy. The derivatives are evaluated at $\widehat{X}_n$.

14.5 Multivariate Itô formula

Let $X_t \in \mathbb{R}^d$ satisfy

$$\mathrm{d}X_t^i = a_i(t, X_t)\,\mathrm{d}t + \sum_{k=1}^{m} b_{ik}(t, X_t)\,\mathrm{d}W_t^k,$$

with independent driving Brownian motions. For $f \in C^{1,2}([0,T] \times \mathbb{R}^d)$,

$$\begin{aligned}\mathrm{d}f(t, X_t) = \left[f_t + \sum_{i=1}^{d} a_i f_{x_i} + \frac{1}{2} \sum_{i,j=1}^{d} \left(\sum_{k=1}^{m} b_{ik} b_{jk} \right) f_{x_i x_j} \right] \mathrm{d}t \\ + \sum_{k=1}^{m} \left(\sum_{i=1}^{d} b_{ik} f_{x_i} \right) \mathrm{d}W_t^k.\end{aligned}$$

Writing $B = (b_{ik})$, the second-order operator is $(1/2)\sum_{ij}(BB^\top)_{ij}\partial_{x_i x_j}$, which directly links the Hessian correction to the covariance matrix. High-order multivariate schemes require approximating iterated stochastic integrals, including cross integrals (Lévy areas); these cannot in general be reduced to independent scalar Milstein corrections.

14.6 A detailed Itô–Taylor expansion

For a time-homogeneous scalar SDE, the integral representation over $[t, t+h]$ is

$$X_{t+h} = X_t + \int_t^{t+h} a(X_s)\,\mathrm{d}s + \int_t^{t+h} b(X_s)\,\mathrm{d}W_s.$$

Apply Itô's formula separately to $a(X_s)$ and $b(X_s)$:

$$a(X_s) = a(X_t) + \int_t^s (L^0 a)(X_u)\,\mathrm{d}u + \int_t^s (L^1 a)(X_u)\,\mathrm{d}W_u,$$
$$b(X_s) = b(X_t) + \int_t^s (L^0 b)(X_u)\,\mathrm{d}u + \int_t^s (L^1 b)(X_u)\,\mathrm{d}W_u.$$

Substituting into the original equation creates iterated integrals. Introduce

$$I_{00} = \int_t^{t+h}\int_t^s \mathrm{d}u\,\mathrm{d}s = \frac{h^2}{2},$$
$$I_{11} = \int_t^{t+h}\int_t^s \mathrm{d}W_u\,\mathrm{d}W_s = \frac{(\Delta W)^2 - h}{2},$$
$$I_{10} = \int_t^{t+h}\int_t^s \mathrm{d}W_u\,\mathrm{d}s,$$
$$I_{01} = \int_t^{t+h}\int_t^s \mathrm{d}u\,\mathrm{d}W_s.$$

Since $I_{10} + I_{01} = h\Delta W$, replacing the two mixed integrals by $h\Delta W/2$ in the appropriate weak expansion leads to the simplified second-order scheme seen in the lecture. This substitution is not a pathwise identity. Indeed I_{10} and I_{01} are different stochastic integrals, so higher *strong* order requires accurate treatment of their joint distribution.

14.7 Differential operators in higher dimensions

For vector-valued $X_t \in \mathbb{R}^d$ driven by m Brownian motions, define

$$L^0 = \partial_t + \sum_{i=1}^d a_i \partial_{x_i} + \frac{1}{2}\sum_{i,j=1}^d (BB^\top)_{ij}\partial_{x_i x_j}, \qquad L^k = \sum_{i=1}^d b_{ik}\partial_{x_i}.$$

An iterated stochastic Taylor series applies these operators to coordinate functions, drift coefficients, and diffusion coefficients. Terms involving $L^k b_{ij}$ measure how the jth diffusion direction changes along the kth Brownian direction. When these derivatives do not commute, the resulting approximation contains mixed stochastic areas and cannot be reconstructed from Brownian increments alone without additional random variables or structural simplifications. This is an important conceptual distinction between scalar Milstein and genuinely multidimensional strong simulation.

14.8 The second-order multivariate approximation in the lecture

For the vector SDE introduced earlier, use the operators $L^0, L^1, \ldots, L^m$. A simplified weak-order-two update for coordinate i is expressed in the lecture in the form

$$\begin{aligned}\widehat{X}_i(t+h) = \widehat{X}_i(t) + a_i h + \sum_{k=1}^m b_{ik}\Delta W_k + \frac{1}{2}(L^0 a_i)h^2 \\ + \frac{1}{2}\sum_{k=1}^m \left(L^k a_i + L^0 b_{ik}\right)\Delta W_k h \\ + \frac{1}{2}\sum_{j=1}^m\sum_{k=1}^m (L^j b_{ik})(\Delta W_j \Delta W_k - V_{jk}).\end{aligned}$$

All coefficients and their operator derivatives are evaluated at the current approximation $(t, \widehat{X}(t))$. The random matrix V in a common weak Taylor construction has $V_{jj} = h$, and for $j < k$ one chooses independent Rademacher values $V_{jk} = \pm h$ with equal probability, setting $V_{kj} = -V_{jk}$. The supplied Lecture 14 slide instead writes $V_{kj} = V_{jk}$, a symmetry convention that differs from the familiar antisymmetric weak-Taylor surrogate. Because different stochastic Taylor schemes can require different auxiliary random matrices, a numerical implementation must verify the corresponding moment conditions and the intended convention rather than combining one convention with another schemeś coefficients. Importantly, these supplementary signs are not the actual Lévy areas. They are random surrogates selected so that relevant low-order moments of the numerical increment match those of the exact SDE to the required weak accuracy.

The scalar version with $m = 1$ has $V_{11} = h$ and recovers the familiar factor $(\Delta W)^2 - h$. The vector version exposes the main additional complexity: distinct noise directions contribute mixed products whose distributional treatment matters for convergence. In implementation, the Brownian increments must be sampled consistently with the covariance of the driving Brownian motions; correlated Brownian drivers should be whitened or represented through a suitable factor matrix before this formula is used.

Lecture 15

Extrapolation, Discretization Bias, and Quasi-Monte Carlo

15.1 Bias and statistical error are different

When an SDE must be discretized, the Monte Carlo estimator uses simulated terminal values $\widehat{X}_T^{(h)}$. Its total estimation error can be decomposed as

$$\widehat{\theta}_{n,h} - \theta = \left(\widehat{\theta}_{n,h} - \mathbb{E}[f(\widehat{X}_T^{(h)})]\right) + \left(\mathbb{E}[f(\widehat{X}_T^{(h)})] - \mathbb{E}[f(X_T)]\right).$$

The first term is statistical Monte Carlo noise, usually $O_{\mathbb{P}}(n^{-1/2})$. The second is discretization bias, often $O(h^p)$ for weak order p. Increasing n cannot remove a fixed step-size bias, and reducing h does not by itself reduce the sampling variance.

15.2 Richardson extrapolation

Suppose a weak approximation satisfies a more detailed expansion

$$\mathbb{E}[f(\widehat{X}_T^{(h)})] = \theta + ch^p + O(h^{p+1}).$$

For the same model, the half-step approximation satisfies $\mathbb{E}[f(\widehat{X}_T^{(h/2)})] = \theta + c(h/2)^p + O(h^{p+1})$. Eliminate the leading bias via

$$\begin{aligned}\theta_h^{\text{ext}} &= \frac{2^p\mathbb{E}[f(\widehat{X}_T^{(h/2)})] - \mathbb{E}[f(\widehat{X}_T^{(h)})]}{2^p - 1} \\ &= \theta + O(h^{p+1}).\end{aligned}$$

When implementing this with Monte Carlo, coupling the coarse and fine estimates using shared Brownian increments can reduce the variance of their difference. Strong errors and covariance structure matter even when the final objective is weak convergence.

15.3 Quasi-Monte Carlo integration

For $g : [0,1]^d \to \mathbb{R}$ and $\theta = \int_{[0,1]^d} g(u)\,\mathrm{d}u$, crude Monte Carlo samples independently, whereas quasi-Monte Carlo (QMC) uses carefully distributed deterministic points $v_1, \ldots, v_n$:

$$\widehat{\theta}_{\text{QMC}} = \frac{1}{n}\sum_{i=1}^{n} g(v_i).$$

Definition 15.1 (Star discrepancy)**.** For $P_n = \{v_1, \dots, v_n\} \subset [0,1]^d$, define

$$D_n^*(P_n) = \sup_{z \in [0,1]^d} \left| \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{[0,z)}(v_i) - \prod_{j=1}^{d} z_j \right|.$$

The star discrepancy measures how well the empirical point frequencies agree with volume in axis-aligned rectangles anchored at the origin. Low-discrepancy constructions such as Sobol' sequences spread points more evenly than a typical independent random sample in low and moderate effective dimension.

Theorem 15.2 (Koksma–Hlawka inequality)**.** *If g has finite variation $V_{\mathrm{HK}}(g)$ in the Hardy–Krause sense, then*

$$\left| \widehat{\theta}_{\mathrm{QMC}} - \theta \right| \le V_{\mathrm{HK}}(g) D_n^*(P_n).$$

For appropriate low-discrepancy sequences, a typical discrepancy bound is $O((\log n)^d/n)$, which can be better than $n^{-1/2}$ for fixed small d and large n. The bound can be uninformative for nonsmooth payoffs, singular transforms, or high dimension. Randomized QMC adds a random scrambling or shift while preserving favorable spatial distribution, allowing replication-based standard error estimates under an appropriate randomization scheme.

15.4 A transparent derivation of extrapolation weights

Suppose a numerical scheme has weak bias $ch + dh^2 + o(h^2)$. Let M_h and $M_{h/2}$ denote its expected payoff approximations. Then

$$M_h = \theta + ch + dh^2 + o(h^2),$$
$$M_{h/2} = \theta + \frac{c}{2}h + \frac{d}{4}h^2 + o(h^2).$$

Form $2M_{h/2} - M_h$:

$$\begin{aligned} 2M_{h/2} - M_h &= 2\theta + ch + \frac{d}{2}h^2 - \theta - ch - dh^2 + o(h^2) \\ &= \theta - \frac{d}{2}h^2 + o(h^2). \end{aligned}$$

Thus a first-order scheme can become second-order accurate for suitably smooth expectations by combining two mesh levels. If estimating M_h and $M_{h/2}$ by independent Monte Carlo runs, the extrapolated variance is $4\,\mathrm{Var}(\widehat{M}_{h/2}) + \mathrm{Var}(\widehat{M}_h)$, which can be costly. Instead, use a paired construction sharing the Brownian path, so

$$\mathrm{Var}(2\widehat{M}_{h/2} - \widehat{M}_h) = 4\,\mathrm{Var}(\widehat{M}_{h/2}) + \mathrm{Var}(\widehat{M}_h) - 4\,\mathrm{Cov}(\widehat{M}_{h/2}, \widehat{M}_h).$$

Positive coarse/fine covariance lowers the variance, an example of common random numbers in discretization error reduction.

15.5 QMC caveats in high-dimensional path simulation

A simulated Brownian trajectory observed at d time points can be expressed as a deterministic transform of a d-dimensional uniform vector. For example, the usual forward construction is

$$W_{t_j} = \sum_{k=1}^{j} \sqrt{t_k - t_{k-1}}\, \Phi^{-1}(u_k).$$

Although the input dimension equals the number of time steps, the payoff may depend principally on a few broad features of the path. Brownian bridge construction assigns early coordinates to coarse-scale path information, potentially reducing effective dimension for quasi-Monte Carlo. This can improve QMC performance even when the nominal dimension remains large. However, endpoint-heavy constructions are not universally optimal: a payoff controlled by an early barrier crossing may respond differently from a terminal-value payoff. Numerical experimentation with replicated randomized QMC is essential.

Lecture 16

Poisson Processes, Jump Models, and Introduction to PDE Pricing

16.1 A Poisson process and exponential waiting times

Definition 16.1 (Homogeneous Poisson process)**.** A counting process $(N_t)_{t\geq 0}$ with rate $\lambda > 0$ satisfies $N_0 = 0$, has independent stationary increments, and for $s < t$,

$$N_t - N_s \sim \text{Poisson}(\lambda(t-s)).$$

Hence $\mathbb{P}(N_t = k) = e^{-\lambda t}(\lambda t)^k/k!$, $\mathbb{E}[N_t] = \lambda t$, and $\text{Var}(N_t) = \lambda t$. Let T_1 denote the first arrival time. The event $\{T_1 > t\}$ is equivalent to having no arrivals in $(0, t]$, so

$$\mathbb{P}(T_1 > t) = \mathbb{P}(N_t = 0) = e^{-\lambda t},$$
$$F_{T_1}(t) = 1 - e^{-\lambda t}, \qquad f_{T_1}(t) = \lambda e^{-\lambda t}.$$

The inter-arrival durations are independent $\text{Exp}(\lambda)$ random variables. To simulate all jump times before T, generate exponentials $E_j = -\log U_j/\lambda$ and form the partial sums $T_k = \sum_{j=1}^{k} E_j$ until exceeding T. Alternatively, simulate the number of events as $\text{Poisson}(\lambda T)$ and conditional uniform jump times, then sort them.

16.2 Jump-diffusion asset models

A simple jump-diffusion can be written

$$\frac{\mathrm{d}S_t}{S_{t-}} = (\mu - \lambda\kappa)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + (J-1)\,\mathrm{d}N_t,$$

where S_{t-} is the left limit, $J > 0$ is a random gross jump multiplier, and $\kappa = \mathbb{E}[J-1]$. The compensator $-\lambda\kappa$ corrects for the average jump contribution. In a risk-neutral model with no dividends, replace μ by r, with appropriate assumptions that ensure the discounted stock is a martingale. Conditional on independently generated jump times, one can propagate the diffusion exactly between arrivals and multiply the stock by each jump factor at an event. This event-driven construction avoids missing discontinuities on a fixed time grid.

16.3 A first derivation of the Black–Scholes PDE

For a European payoff $H(S_T)$, let $V(t, s)$ denote its smooth model price in the risk-neutral GBM model. Itô's formula yields

$$\mathrm{d}V(t, S_t) = \left(V_t + rS_tV_s + \tfrac{1}{2}\sigma^2 S_t^2 V_{ss}\right)\,\mathrm{d}t + \sigma S_t V_s\,\mathrm{d}W_t.$$

Form the delta-hedged portfolio consisting of one option and short V_s shares. The Brownian fluctuation cancels, leaving instantaneous drift

$$\mathrm{d}V - V_s\,\mathrm{d}S = \left(V_t + \tfrac{1}{2}\sigma^2 S^2 V_{ss}\right)\,\mathrm{d}t.$$

The portfolio value is $V - SV_s$. Absence of arbitrage requires its riskless drift to be $r(V - SV_s)\,\mathrm{d}t$. Equating terms and rearranging produces

$$V_t + \tfrac{1}{2}\sigma^2 s^2 V_{ss} + rsV_s - rV = 0, \qquad V(T, s) = H(s).$$

The PDE is a *backward terminal-value problem*: its terminal payoff is known, and the objective is to infer earlier values. Numerical methods solve this PDE when a convenient closed form does not exist.

16.4 The generator and its probabilistic meaning

For a diffusion with drift rs and variance rate $\sigma^2 s^2$, the infinitesimal generator acting on smooth functions is

$$(\mathcal{A}f)(s) = rsf'(s) + \tfrac{1}{2}\sigma^2 s^2 f''(s).$$

The pricing equation is $V_t + \mathcal{A}V - rV = 0$. It expresses the same risk-neutral discounting structure as $V(t, s) = \mathbb{E}^{\mathbb{Q}}_{t,s}[e^{-r(T-t)}H(S_T)]$ under conditions of Feynman–Kac. The relationship between stochastic processes, conditional expectations, and parabolic PDEs is one of the organizing ideas of numerical finance.

16.5 Conditional jump simulation and compound Poisson losses

Suppose a jump process has independent jump sizes $Y_1, Y_2, \ldots$ and rate-λ counting process N_t. Define the compound Poisson process

$$J_t = \sum_{k=1}^{N_t} Y_k.$$

If $\mathbb{E}[|Y_1|] < \infty$, conditioning on N_t gives

$$\begin{aligned}\mathbb{E}[J_t] &= \mathbb{E}\left[\mathbb{E}\left[\sum_{k=1}^{N_t} Y_k \mid N_t\right]\right] \\ &= \mathbb{E}[N_t]\mathbb{E}[Y_1] = \lambda t \mathbb{E}[Y_1].\end{aligned}$$

If $\mathbb{E}[Y_1^2] < \infty$, the law of total variance gives

$$\begin{aligned}\mathrm{Var}(J_t) &= \mathbb{E}[N_t\,\mathrm{Var}(Y_1)] + \mathrm{Var}(N_t\mathbb{E}[Y_1]) \\ &= \lambda t\,\mathrm{Var}(Y_1) + \lambda t(\mathbb{E}[Y_1])^2 \\ &= \lambda t \mathbb{E}[Y_1^2].\end{aligned}$$

These identities are direct tests for a jump simulator and explain why jumps create additional variability beyond the Brownian diffusion term.

16.6 Risk-neutral jump compensation

Consider a stock with proportional random jumps, interpreted through

$$\frac{\mathrm{d}S_t}{S_{t-}} = b\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + (Y-1)\,\mathrm{d}N_t.$$

Over an infinitesimal period $\mathrm{d}t$, the expected jump contribution to $\mathrm{d}S_t/S_{t-}$ equals $\lambda\mathbb{E}[Y-1]\,\mathrm{d}t$. Therefore the total expected percentage drift is $b+\lambda\mathbb{E}[Y-1]$. To make the undiscounted risk-neutral stock have conditional expected return r, choose

$$b = r - \lambda\mathbb{E}[Y-1].$$

If dividends are paid continuously at rate q, replace r by $r-q$. Missing the jump compensation changes forward prices and invalidates the discounted-martingale property. Exact event-time simulation can combine this compensated deterministic drift with Brownian innovations and multiplicative jump factors.

16.7 PDE versus integro-differential equations

A pure diffusion has a local generator containing only derivatives. A jump-diffusion's generator also contains a nonlocal expectation over post-jump states. With jump multiplier Y and intensity λ, one schematic risk-neutral generator is

$$(\mathcal{A}f)(s) = (r-\lambda\kappa)sf'(s) + \frac{1}{2}\sigma^2 s^2 f''(s) + \lambda\mathbb{E}[f(sY)-f(s)].$$

Accordingly, an option value generally satisfies a partial integro-differential equation (PIDE)

$$V_t + \mathcal{A}V - rV = 0$$

instead of a purely differential Black–Scholes PDE. Numerically, the integral introduces nonlocal coupling between stock-grid nodes, illustrating why simulation is particularly attractive for certain jump and path-dependent models.

Lecture 17

Finite Differences for the Black–Scholes PDE

17.1 From option valuation to a parabolic PDE

For a non-dividend-paying stock under risk-neutral geometric Brownian motion, the value $V(s,t)$ of a smooth European contingent claim solves

$$V_t + \frac{1}{2}\sigma^2 s^2 V_{ss} + rsV_s - rV = 0, \qquad 0 \le t < T,$$

with terminal condition $V(s,T) = H(s)$. The PDE is backward in calendar time. To write it as a forward evolution equation in time-to-maturity, introduce $\tau = T - t$ and $v(s,\tau) = V(s, T-\tau)$. The chain rule gives $v_\tau = -V_t$, so

$$v_\tau = \frac{1}{2}\sigma^2 s^2 v_{ss} + rsv_s - rv, \qquad v(s,0) = H(s).$$

We work on a truncated spatial interval $0 = s_0 < s_1 < \cdots < s_M = s_{\max}$ and time grid $\tau_n = n\Delta\tau$, $n = 0, \dots, N$, with $N\Delta\tau = T$. Let $v_i^n \approx v(s_i, \tau_n)$. The design of a reliable finite-difference scheme requires two independent ingredients: consistent approximations of derivatives and appropriate boundary conditions.

17.2 Finite-difference formulas from Taylor expansions

For simplicity begin with an equally spaced spatial grid $s_{i+1} - s_i = h$. Expand a smooth function f about s_i:

$$f(s_i + h) = f_i + hf_i' + \frac{h^2}{2}f_i'' + \frac{h^3}{6}f_i''' + O(h^4),$$
$$f(s_i - h) = f_i - hf_i' + \frac{h^2}{2}f_i'' - \frac{h^3}{6}f_i''' + O(h^4).$$

Subtract the expansions, divide by $2h$, and obtain the central first derivative:

$$f'(s_i) = \frac{f_{i+1} - f_{i-1}}{2h} + O(h^2).$$

Alternatively, the one-sided first differences are

$$f'(s_i) = \frac{f_{i+1} - f_i}{h} + O(h), \qquad f'(s_i) = \frac{f_i - f_{i-1}}{h} + O(h).$$

Adding the Taylor expansions gives

$$f_{i+1} + f_{i-1} = 2f_i + h^2 f_i'' + O(h^4),$$
$$f''(s_i) = \frac{f_{i+1} - 2f_i + f_{i-1}}{h^2} + O(h^2).$$

For a nonuniform grid, write $h_- = s_i - s_{i-1}$ and $h_+ = s_{i+1} - s_i$. A useful local approximation is

$$v_{ss}(s_i) \approx \frac{2}{h_- + h_+} \left(\frac{v_{i+1} - v_i}{h_+} - \frac{v_i - v_{i-1}}{h_-} \right).$$

This formula is consistent for smooth functions as the grid is refined; the precise global order depends on mesh regularity.

17.3 Explicit time stepping

Define local coefficients for an interior node by

$$a_i = \frac{1}{2}\sigma^2 s_i^2, \qquad b_i = r s_i.$$

On a uniform grid with central differences, write the spatial operator in the form

$$(\mathcal{L}_h v)_i = \alpha_i (v_{i-1} - v_i) + \beta_i (v_{i+1} - v_i) - r v_i,$$

where

$$\alpha_i = \frac{a_i}{h^2} - \frac{b_i}{2h}, \qquad \beta_i = \frac{a_i}{h^2} + \frac{b_i}{2h}.$$

Indeed,

$$\alpha_i (v_{i-1} - v_i) + \beta_i (v_{i+1} - v_i) = \frac{a_i}{h^2}(v_{i-1} - 2v_i + v_{i+1}) + \frac{b_i}{2h}(v_{i+1} - v_{i-1}).$$

A forward difference in τ gives $v_i^{n+1} = v_i^n + \Delta\tau (\mathcal{L}_h v^n)_i$, or

$$v_i^{n+1} = \Delta\tau \alpha_i v_{i-1}^n + \left[1 - \Delta\tau(\alpha_i + \beta_i + r)\right] v_i^n + \Delta\tau \beta_i v_{i+1}^n.$$

When all three coefficients are nonnegative and sum to $1 - r\Delta\tau$ with $r \geq 0$, the update has a positivity-preserving structure. A sufficient local step restriction is

$$\Delta\tau \leq \frac{1}{\max_i (\alpha_i + \beta_i + r)}$$

provided $\alpha_i, \beta_i \geq 0$ for all interior nodes.

17.4 Initial and boundary data

The terminal payoff becomes the numerical initial condition $v_i^0 = H(s_i)$. For the Black–Scholes European call, the natural asymptotic boundaries are

$$v(0,\tau) = 0, \qquad v(s,\tau) \sim s - Ke^{-r\tau} \quad (s \to \infty).$$

For the European put,

$$v(0,\tau) = Ke^{-r\tau}, \qquad v(s,\tau) \to 0 \quad (s \to \infty).$$

These conditions follow from the probabilistic payoff and no-arbitrage bounds and are more accurate at a finite upper boundary than arbitrarily holding the upper-node value fixed. At $s = 0$, the PDE reduces to $v_\tau = -rv$ for payoffs consistent with an absorbing zero stock price. An explicit lower-boundary update is $v_0^{n+1} = (1 - r\Delta\tau)v_0^n$; exact discounting or a higher-order discretization is also possible.

17.5 Consistency obtained by inserting the exact solution

Consider the differential operator

$$(\mathcal{L}v)(s,\tau) = a(s)v_{ss}(s,\tau) + b(s)v_s(s,\tau) - rv(s,\tau).$$

Let v be a sufficiently smooth exact solution of $v_\tau = \mathcal{L}v$. Evaluate the numerical explicit residual at a grid point:

$$R_i^n = \frac{v(s_i,\tau_{n+1}) - v(s_i,\tau_n)}{\Delta\tau} - (\mathcal{L}_h v(\cdot,\tau_n))_i.$$

The time Taylor expansion yields

$$\frac{v(s_i,\tau_{n+1}) - v(s_i,\tau_n)}{\Delta\tau} = v_\tau(s_i,\tau_n) + \frac{\Delta\tau}{2}v_{\tau\tau}(s_i,\tau_n) + O(\Delta\tau^2).$$

With centered spatial derivatives on a uniform grid,

$$\mathcal{L}_h v = \mathcal{L}v + O(h^2).$$

Substituting $v_\tau = \mathcal{L}v$ gives

$$R_i^n = O(\Delta\tau) + O(h^2).$$

This is the consistency error in the differential equation. It does not prove the numerical solution converges, because errors can amplify when the scheme is iterated. A separate stability argument is necessary.

17.6 Why the high-stock boundary must be handled carefully

For a call with terminal payoff $(s-K)^+$, the asymptotic form $v(s,\tau) \sim s - Ke^{-r\tau}$ is determined by the event that a sufficiently large stock price almost surely leaves the call in the money at maturity. More precisely, the Black–Scholes formula has $d_1, d_2 \to +\infty$ as $s \to \infty$, so $\Phi(d_1), \Phi(d_2) \to 1$ and

$$v(s,\tau) = s\Phi(d_1) - Ke^{-r\tau}\Phi(d_2) \sim s - Ke^{-r\tau}.$$

Imposing the constant upper-node condition $v(s_{\max}, \tau) = v(s_{\max}, 0) = s_{\max} - K$ ignores the discounting of the strike and can introduce a bias even when the spatial mesh is extremely fine. Choosing a more accurate asymptotic condition is therefore a distinct numerical issue from refining the derivative stencil.

17.7 Nonuniform-grid first derivatives

For $h_- = s_i - s_{i-1}$ and $h_+ = s_{i+1} - s_i$, a three-point nonuniform derivative approximation can be obtained by eliminating the second derivative from asymmetric Taylor expansions. The result is

$$f'(s_i) \approx -\frac{h_+}{h_-(h_- + h_+)} f_{i-1} + \frac{h_+ - h_-}{h_- h_+} f_i + \frac{h_-}{h_+(h_- + h_+)} f_{i+1}.$$

If $h_- = h_+ = h$, it reduces to $(f_{i+1} - f_{i-1})/(2h)$. In practice, clustering points near a strike can increase local pricing accuracy without multiplying the total number of nodes as aggressively as a globally fine uniform grid.

Lecture 18

Stability, Upwinding, Implicit Schemes, and Crank–Nicolson

18.1 Monotonicity and upwind differencing

Central differencing of the drift term is second-order in space but can produce $\alpha_i < 0$ where diffusion is weak relative to the drift. Such negative coefficients may cause oscillations, loss of positivity, or poor stability even when a formally higher-order truncation error is advertised. For $b_i > 0$, a forward first difference for v_s gives

$$\alpha_i^{\text{fwd}} = \frac{a_i}{h^2}, \qquad \beta_i^{\text{fwd}} = \frac{a_i}{h^2} + \frac{b_i}{h},$$

whereas a backward first difference gives

$$\alpha_i^{\text{bwd}} = \frac{a_i}{h^2} - \frac{b_i}{h}, \qquad \beta_i^{\text{bwd}} = \frac{a_i}{h^2}.$$

For the *forward-in-τ* parabolic pricing PDE, select a first-derivative stencil that gives nonnegative off-diagonal coefficients; which one is stable depends on the PDE sign convention and whether one views the convection term in conservative form. On the above grid with $b_i \geq 0$, the displayed forward formula preserves nonnegative off-diagonal weights. One may use central differences whenever $\alpha_i, \beta_i \geq 0$ and switch to a one-sided formula only where the condition fails. This is the scheme-selection idea discussed under upstream weighting in the lectures.

18.2 Matrix formulation

For interior nodes, collect the spatial discretization into a matrix A_h such that $(A_h v)_i = (\mathcal{L}_h v)_i$. Define $M = -\Delta\tau A_h$, with boundary rows adjusted to impose the chosen conditions. The explicit scheme becomes

$$v^{n+1} = (I - M)v^n.$$

At an interior row the nonzero coefficients of M are

$$M_{i,i-1} = -\Delta\tau\alpha_i, \quad M_{i,i} = \Delta\tau(\alpha_i + \beta_i + r), \quad M_{i,i+1} = -\Delta\tau\beta_i.$$

The matrix is tridiagonal, a computational advantage: linear systems can be solved in $O(M)$ work using the Thomas algorithm when suitable pivoting conditions hold.

18.3 Fully implicit Euler method

Evaluate the spatial right-hand side at the new time level:

$$\frac{v_i^{n+1} - v_i^n}{\Delta\tau} = \alpha_i(v_{i-1}^{n+1} - v_i^{n+1}) + \beta_i(v_{i+1}^{n+1} - v_i^{n+1}) - rv_i^{n+1}.$$

Multiplying by $\Delta\tau$ and moving new unknowns to the left produces

$$(I + M)v^{n+1} = v^n.$$

Under standard monotonicity conditions, the implicit method is stable for all positive time steps even though excessively large steps degrade accuracy. In the constant-coefficient scalar test equation $y' = -\lambda y$, $\lambda \geq 0$, implicit Euler has amplification factor $(1 + \lambda\Delta\tau)^{-1}$, whose magnitude never exceeds one.

18.4 Crank–Nicolson as trapezoidal time integration

Average the spatial operator at the old and new times:

$$\frac{v^{n+1} - v^n}{\Delta\tau} = \frac{1}{2}A_h(v^{n+1} + v^n).$$

Rearrangement gives

$$\left(I - \frac{\Delta\tau}{2}A_h\right)v^{n+1} = \left(I + \frac{\Delta\tau}{2}A_h\right)v^n,$$
$$\left(I + \frac{1}{2}M\right)v^{n+1} = \left(I - \frac{1}{2}M\right)v^n.$$

For a smooth solution on a uniform spatial mesh, Crank–Nicolson has local discretization errors of order $O(\Delta\tau^2 + h^2)$ in the standard global consistency sense, whereas forward and backward Euler are first-order in time. For nonsmooth option payoffs, startup damping (for example, two implicit half-steps) can reduce the initial spurious oscillations associated with Crank–Nicolson.

18.5 A lower-boundary Crank–Nicolson equation

At $s = 0$ the PDE reduces to $\partial_\tau v = -rv$. Applying the trapezoidal method gives

$$\frac{v_0^{n+1} - v_0^n}{\Delta\tau} = -\frac{r}{2}(v_0^{n+1} + v_0^n),$$
$$\left(1 + \frac{r\Delta\tau}{2}\right)v_0^{n+1} = \left(1 - \frac{r\Delta\tau}{2}\right)v_0^n.$$

Accordingly, the first row of $M/2$ has diagonal entry $r\Delta\tau/2$ and no off-diagonal entries. This is equivalent to setting $\alpha_0 = \beta_0 = 0$ while retaining the discount term.

18.6 Positivity and a discrete maximum principle

Suppose the explicit coefficients are nonnegative:

$$A_i = \Delta\tau\alpha_i \geq 0, \quad B_i = 1 - \Delta\tau(\alpha_i + \beta_i + r) \geq 0, \quad C_i = \Delta\tau\beta_i \geq 0.$$

For $r \geq 0$, they satisfy $A_i + B_i + C_i = 1 - r\Delta\tau \leq 1$. If all values at time n are nonnegative, then $v_i^{n+1} = A_i v_{i-1}^n + B_i v_i^n + C_i v_{i+1}^n \geq 0$. Hence, under suitable nonnegative boundary conditions, the scheme preserves nonnegative option values.

Similarly, if the two solutions v and w have ordered initial and boundary values, and use the same coefficients, then $v^n \leq w^n$ componentwise implies $v^{n+1} \leq w^{n+1}$ componentwise. This discrete comparison principle mirrors the continuous parabolic PDE's comparison principle. Negative coefficients destroy this simple reasoning; an apparently small numerical truncation error may coexist with negative computed option prices or oscillatory Greeks.

18.7 Fourier stability for a constant-coefficient heat equation

A standard model problem is $u_\tau = \kappa u_{xx}$ with $\kappa > 0$. The forward-time, centered-space scheme on a uniform grid is

$$u_i^{n+1} = u_i^n + \nu(u_{i-1}^n - 2u_i^n + u_{i+1}^n), \qquad \nu = \frac{\kappa\Delta\tau}{h^2}.$$

For a Fourier error mode $e_i^n = G^n e^{\mathrm{i}i\xi}$, substitute into the scheme:

$$\begin{aligned} G &= 1 + \nu(e^{-\mathrm{i}\xi} - 2 + e^{\mathrm{i}\xi}) \\ &= 1 - 2\nu(1 - \cos\xi) = 1 - 4\nu\sin^2(\xi/2). \end{aligned}$$

For stability it is necessary and sufficient on the infinite/periodic uniform grid that $|G| \leq 1$ for every ξ. Since $0 \leq \sin^2(\xi/2) \leq 1$, this holds exactly when $0 \leq \nu \leq 1/2$. Consequently,

$$\Delta\tau \leq \frac{h^2}{2\kappa}.$$

An explicit diffusion scheme therefore requires a time step of order h^2 as the spatial mesh is refined, which can be costly. Fully implicit and Crank–Nicolson schemes avoid this particular stability restriction, although they require solving linear systems.

Lecture 19

Convergence, Dividends, and Market Calibration

19.1 Stability, consistency, and convergence

A numerical approximation v_i^n is *convergent* if v_i^n approaches the true solution evaluated at the corresponding mesh points as $\Delta\tau$ and the maximum spatial mesh size tend to zero. Consistency controls the local error made by inserting a smooth exact solution into the scheme. Stability controls the propagation of perturbations and rounding errors from one time step to the next. For well-posed linear initial-value problems, the Lax equivalence framework relates a consistent scheme's stability to its convergence, under the relevant assumptions.

For a one-step linear method $v^{n+1} = B_h v^n$, repeated application yields $v^n = B_h^n v^0$. Large powers of B_h magnify disturbances. In a positivity-preserving explicit update, nonnegative rows and an appropriate norm bound can help control these powers. For a stable implicit discretization, the matrix inverse $(I + M)^{-1}$ damps high-frequency components. Stability alone is not accuracy: truncating the stock domain too narrowly or using an inappropriate upper boundary can dominate the PDE discretization error.

19.2 Choice of computational domain and mesh

The stock interval $[0, s_{\max}]$ approximates an unbounded state space. If $s_{\max}$ is too small, boundary conditions contaminate prices near the current stock level. An effective procedure is to enlarge $s_{\max}$ until the value of the target option changes by less than a specified tolerance. Grid refinement in both space and time can then distinguish boundary truncation error from finite-difference error.

Use finer spatial nodes near strikes, barriers, or other payoff kinks. Time steps can be shortened around sudden changes in exercise constraints or around dividend dates. Numerical Greeks are especially sensitive to grid spacing and payoff nonsmoothness, so convergence checks should include sensitivities as well as the option price.

19.3 Continuous dividend yields

If the stock pays a constant continuous dividend yield q, its risk-neutral ex-dividend dynamics are

$$\mathrm{d}S_t = (r - q)S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^{\mathbb{Q}}.$$

A self-financing position in the stock receives the dividend stream $qS_t\,\mathrm{d}t$. Delta-hedging the option and accounting for that income replaces rS by $(r - q)S$ in the PDE:

$$V_t + \frac{1}{2}\sigma^2 s^2 V_{ss} + (r - q)sV_s - rV = 0.$$

For European calls and puts, the put–call parity identity becomes

$$C_0 - P_0 = S_0 e^{-qT} - K e^{-rT}.$$

Ignoring dividends produces systematically wrong forward levels, option values, and hedge deltas.

19.4 Black–Scholes closed forms and implied volatility

Let $\tau = T - t > 0$, with $s > 0$, $K > 0$, and $\sigma > 0$. For a European call under a constant dividend yield q, the Black–Scholes formula is

$$\begin{aligned} C(t, s; K, T, \sigma) &= s e^{-q\tau} \Phi(d_1) - K e^{-r\tau} \Phi(d_2), \\ d_1 &= \frac{\log(s/K) + (r - q + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \\ d_2 &= d_1 - \sigma\sqrt{\tau}. \end{aligned}$$

The corresponding put is $P = K e^{-r\tau} \Phi(-d_2) - s e^{-q\tau} \Phi(-d_1)$. For an observed option market price C^{mkt}, its Black–Scholes implied volatility σ_{imp} solves

$$C(t, s; K, T, \sigma_{\mathrm{imp}}) = C^{\mathrm{mkt}}.$$

This is a numerical root-finding problem. Since call vega satisfies

$$\frac{\partial C}{\partial \sigma} = s e^{-q\tau} \varphi(d_1) \sqrt{\tau} > 0,$$

the theoretical call price is strictly increasing in volatility for positive maturity and a nondegenerate underlying. Newton's method is

$$\sigma_{k+1} = \sigma_k - \frac{C(\sigma_k) - C^{\mathrm{mkt}}}{\partial_\sigma C(\sigma_k)}.$$

Bracketing and careful handling of low-vega deep in/out-of-the-money contracts improve numerical robustness.

19.5 The volatility smile and fitting model parameters

If the Black–Scholes model with constant σ were exactly correct across all strikes and maturities, every liquid European option on the same underlying would imply the same volatility. In practice $\sigma_{\mathrm{imp}}(K, T)$ varies with both strike and maturity, producing smiles, skews, and a volatility surface. This motivates more expressive models with local volatility, stochastic volatility, jumps, or mixtures.

Given market quotes C_l^{mkt} and a parameterized model price $C_l(\theta)$, calibration estimates

$$\widehat{\theta} \in \arg\min_{\theta \in \Theta} \sum_{l=1}^{m} w_l \big(C_l(\theta) - C_l^{\mathrm{mkt}}\big)^2,$$

where $w_l > 0$ are chosen error weights. The objective is typically nonlinear, and reliable calibration requires constraints, sensible initial values, sufficient data quality, and a consistent numerical pricing algorithm. Models must fit market instruments well enough to be useful for marking-to-market and hedging, but a fit at observed strikes does not guarantee accurate extrapolation to unquoted payoffs.

19.6 The theta family of time-stepping schemes

Forward Euler, backward Euler, and Crank–Nicolson can be unified using a weight $\vartheta \in [0, 1]$:

$$\frac{v^{n+1} - v^n}{\Delta\tau} = (1 - \vartheta)A_h v^n + \vartheta A_h v^{n+1}.$$

Equivalently,

$$(I - \vartheta\Delta\tau A_h)v^{n+1} = (I + (1 - \vartheta)\Delta\tau A_h)v^n.$$

The choices $\vartheta = 0$, 1, and $1/2$ produce explicit Euler, implicit Euler, and Crank–Nicolson. For the scalar test equation $y' = -\lambda y$, the amplification factor is

$$G(z) = \frac{1 - (1 - \vartheta)z}{1 + \vartheta z}, \qquad z = \lambda\Delta\tau \geq 0.$$

For $\vartheta \geq 1/2$, $|G(z)| \leq 1$ for every $z \geq 0$. Crank–Nicolson is therefore stable on the negative real axis but not strongly damping for $z \to \infty$: $G(z) \to -1$ when $\vartheta = 1/2$. Implicit Euler has $G(z) \to 0$ and therefore damps stiff, rapidly oscillating modes, explaining why initial implicit steps can improve a Crank–Nicolson option solver near a nonsmooth payoff.

19.7 Black–Scholes parity under continuous dividends

The put–call parity relation with dividend yield q is derived by comparing a call-minus-put portfolio to a prepaid forward. At maturity, the long call and short put pay $S_T - K$. A prepaid forward delivering one share at T costs $S_0 e^{-qT}$ under the constant-yield model; a zero-coupon bond paying K at T costs Ke^{-rT}. By no arbitrage,

$$C_0 - P_0 = S_0 e^{-qT} - Ke^{-rT}.$$

This also identifies the maturity-T forward price

$$F_{0,T} = S_0 e^{(r-q)T}.$$

The forward relationship is useful when formulating option pricing and implied volatility in terms of log-moneyness $\log(K/F_{0,T})$ rather than raw strikes.

Lecture 20

Calibration and Least-Squares Monte Carlo for American Options

20.1 The optimal stopping formulation

A European claim must be exercised at its prescribed maturity. An American claim permits exercise at a stopping time τ chosen using only information available up to that time. For a discounted nonnegative exercise payoff $g(S_t, t)$, its time-zero value is

$$V_0^{\mathrm{A}} = \sup_{\tau \in \mathcal{T}_{[0,T]}} \mathbb{E}^{\mathbb{Q}} \left[e^{-r\tau} g(S_\tau, \tau) \right],$$

where $\mathcal{T}_{[0,T]}$ is the set of admissible stopping times taking values in $[0, T]$. Restricting exercise to dates $0 = t_0 < \cdots < t_N = T$ gives a Bermudan approximation; its price is no greater than the fully American value because fewer exercise decisions are permitted.

20.2 Why naive pathwise exercise is biased

Suppose one first simulates an entire future trajectory and then retrospectively exercises at the most favorable date on that path. The rule

$$\max_{0 \le j \le N} e^{-rt_j} g(S_{t_j}, t_j)$$

uses future prices not available to the holder at earlier times. It is therefore an *anticipating* or clairvoyant maximum, rather than an admissible stopping strategy, and generally yields an upper-biased value. The critical numerical task is to estimate the *conditional continuation value* using information available at the decision date.

20.3 The Snell-envelope recursion

Set $G_j = g(S_{t_j}, t_j)$ and define the value process V_j through backward induction:

$$\begin{aligned} V_N &= G_N, \\ V_j &= \max \left\{ G_j, e^{-r(t_{j+1} - t_j)} \mathbb{E}^{\mathbb{Q}}[V_{j+1} \mid \mathcal{F}_{t_j}] \right\}, \qquad j = N-1, \ldots, 0. \end{aligned}$$

The conditional expectation is the continuation value of not exercising at t_j. The holder exercises when the immediate payoff is at least the continuation value. This is the discrete-time Snell envelope, the smallest supermartingale dominating the discounted payoff process.

20.4 Longstaff–Schwartz conditional regression

Simulate independent stock trajectories $S_{i,j}$ for paths $i = 1, \dots, n$ and times $j = 0, \dots, N$. For each exercise date, approximate continuation values using a finite set of basis functions $\psi_0, \dots, \psi_M$:

$$C_j(s) \approx \sum_{k=0}^{M} \beta_{j,k} \psi_k(s).$$

For a plain American put, one may use monomials, Laguerre polynomials, or other basis functions in s. The regression at time t_j fits suitably discounted realized future cash flows on current state features:

$$\widehat{\beta}_j \in \arg\min_{\beta} \sum_{i \in \mathcal{I}_j} \left(Y_{i,j} - \sum_{k=0}^{M} \beta_k \psi_k(S_{i,j}) \right)^2 .$$

Here $\mathcal{I}_j$ is often the subset of in-the-money simulated paths, and $Y_{i,j}$ is the cash flow generated by the existing future exercise policy, discounted back to t_j. The estimated continuation function is $\widehat{C}_j(s) = \sum_k \widehat{\beta}_{j,k} \psi_k(s)$.

Algorithm 20.1 (Least-squares Monte Carlo)**.** Simulate n independent complete paths and initialize every path's exercise time at T with its terminal payoff. For $j = N-1, \dots, 1$, discount each path's currently designated future cash flow to t_j. Regress these discounted cash flows on basis functions of the current state, preferably using in-the-money paths. For each eligible path, compare immediate exercise $g(S_{i,j}, t_j)$ to estimated continuation $\widehat{C}_j(S_{i,j})$. If immediate exercise is larger, replace that path's future exercise time by t_j. After processing all dates, average the realized discounted cash flows from the resulting stopping policy. For evaluation, preferably apply the fitted exercise policy to an independent set of test paths.

20.5 Bias, basis selection, and state dependence

Using the same paths for estimating regression coefficients and evaluating the resulting stopping rule can introduce in-sample bias. An independent training/test split removes the most direct look-ahead from in-sample policy evaluation; such an out-of-sample stopping-policy estimate is a lower bound on the true optimal stopping value in expectation, when the policy is admissible. Regression with too few basis functions underfits continuation values; too many relative to the number of in-the-money paths causes unstable estimates. Path-dependent payoffs require additional state variables such as running averages or maxima. Regression provides a flexible method when the option cannot be represented by a tractable low-dimensional PDE.

20.6 Regression as a projection of future cash flows

The continuation value at a decision date t_j is the conditional expectation

$$C_j(X_j) = \mathbb{E}^{\mathbb{Q}}[e^{-r\Delta t} V_{j+1} \mid X_j],$$

where X_j represents the state sufficient for future pricing, possibly containing stock price, volatility, and path statistics. Least-squares Monte Carlo selects a finite-dimensional function space

$$\mathcal{H}_M = \operatorname{span}\{\psi_0, \dots, \psi_M\}$$

and approximates C_j by its L^2 projection onto that space. If $\psi(X)$ is the vector of basis values, the population normal equations are

$$\mathbb{E}[\psi(X_j)\psi(X_j)^\top]\beta_j = \mathbb{E}[\psi(X_j)Y_j],$$

where Y_j is the discounted future cash flow under the current stopping policy. Empirical least squares replaces the expectations by sample averages. This reveals that the method estimates *conditional expectations*, not exercise probabilities directly.

20.7 An explicit two-date stopping example

Consider exercise dates t_1 and $t_2 = T$ and payoff $g(s) = (K - s)^+$. At maturity, $V_2 = g(S_T)$. At t_1, the true continuation value is

$$C_1(s) = e^{-r(T-t_1)}\mathbb{E}^{\mathbb{Q}}[(K - S_T)^+ \mid S_{t_1} = s].$$

Under GBM, this conditional expectation is the European put price with current stock s and remaining maturity $T - t_1$. The optimal Bermudan decision is exercise at t_1 when $(K - s)^+ \geq C_1(s)$. A least-squares approximation instead uses simulated pairs

$$\left(S_{t_1}^{(i)}, e^{-r(T-t_1)}(K - S_T^{(i)})^+\right).$$

The regression response is not the current intrinsic value: it is the realized, suitably discounted *future* cash flow. The current exercise payoff enters only when the regression estimate is compared with the immediate payoff to choose a stopping decision.

20.8 Conditional-policy lower bounds

An estimated exercise strategy can always be evaluated by simulating fresh paths and executing its stopping decisions using current and past states only. Each resulting exercise date $\widehat{\tau}$ is a valid stopping time. Thus

$$\mathbb{E}^{\mathbb{Q}}[e^{-r\widehat{\tau}}g(S_{\widehat{\tau}}, \widehat{\tau})] \leq V_0^{\mathrm{A}}.$$

The fresh-path Monte Carlo estimate of this policy's value is subject to random sampling error, but its population expectation is a lower bound on the optimal American price. This distinction is important: an in-sample training value may not have the same useful lower-bound interpretation if the evaluation procedure selects decisions using information leaked from the training paths.

Lecture 21

American Exercise Properties and the Complementarity PDE

21.1 General ordering properties

Let $V^{\mathrm{A}}(s,t)$ and $V^{\mathrm{E}}(s,t)$ denote American and European values for identical terminal payoffs. Since the holder may exercise immediately,

$$V^{\mathrm{A}}(s,t) \geq g(s,t).$$

Since waiting until expiry is an admissible American exercise strategy,

$$V^{\mathrm{A}}(s,t) \geq V^{\mathrm{E}}(s,t).$$

The difference is the early-exercise premium, which depends on rates, dividends, payoff shape, and the stock dynamics.

Proposition 21.1 (No early exercise for a non-dividend-paying American call)**.** *Assume $r \geq 0$, the stock pays no dividends, and the option is a standard call with strike $K > 0$. Early exercise is never strictly optimal before maturity.*

Proof. Put–call parity and nonnegativity of the European put imply

$$\begin{aligned} C^{\mathrm{E}}(s,t) &= P^{\mathrm{E}}(s,t) + s - Ke^{-r(T-t)} \\ &\geq s - Ke^{-r(T-t)} \geq s - K. \end{aligned}$$

The call price is also nonnegative, hence $C^{\mathrm{E}}(s,t) \geq (s-K)^+$. Holding the European call is at least as valuable as exercising the American call immediately. Because exercise at maturity remains available, the American call has no early-exercise advantage over its European counterpart. □

An American put can benefit from early exercise when deeply in the money, because receiving the strike cash early may be valuable. A dividend-paying call can also favor early exercise when foregone dividends outweigh the benefit of delaying strike payment.

21.2 Continuation and exercise regions

Write the differential operator, under a continuous dividend yield q, as

$$\mathcal{L}V = V_t + \tfrac{1}{2}\sigma^2 s^2 V_{ss} + (r-q)sV_s - rV.$$

Define the continuation region $\mathcal{C} = \{(s,t) : V^{\mathrm{A}}(s,t) > g(s,t)\}$ and exercise region $\mathcal{E} = \{(s,t) : V^{\mathrm{A}}(s,t) = g(s,t)\}$. Inside $\mathcal{C}$, the option is held and behaves locally like a European option, so $\mathcal{L}V^{\mathrm{A}} = 0$. In the exercise region, immediate exercise determines its value, while the superhedging condition gives $\mathcal{L}V^{\mathrm{A}} \leq 0$ in the relevant variational sense.

Theorem 21.2 (American option complementarity problem)**.** *Under sufficient regularity and standard no-arbitrage assumptions, the American value satisfies the obstacle system*

$$\begin{cases} -\mathcal{L}V \geq 0, \\ V - g \geq 0, \\ (V-g)(-\mathcal{L}V) = 0, \\ V(s,T) = g(s,T). \end{cases}$$

Equivalently, $\max\{\mathcal{L}V, g - V\} = 0$.

The first two inequalities require the discounted option value to respect its exercise opportunity and supermartingale property. The product condition enforces that at every point at least one constraint binds: either the holder exercises, or the PDE holds with equality. A classical smooth solution may not exist across the exercise boundary; variational inequalities and viscosity/weak-solution interpretations handle limited regularity.

21.3 Free boundaries and smooth pasting

For an American put on a non-dividend-paying underlying, exercise typically occurs for $s \leq s^*(t)$, where $s^*(t)$ is an unknown exercise boundary. In the stopping region, $V^{\mathrm{A}}(s,t) = K - s$. Under standard regularity for $t < T$, the free boundary satisfies value matching and smooth pasting:

$$V^{\mathrm{A}}(s^*(t),t) = K - s^*(t), \qquad V_s^{\mathrm{A}}(s^*(t),t) = -1.$$

The boundary is part of the solution, which distinguishes the American PDE from a standard linear terminal-boundary problem.

21.4 The American option obstacle as a variational inequality

For a standard European claim the value solves $\mathcal{L}V = 0$ in the full interior of its domain. For an American claim, the holder can terminate the contract, and the exercise payoff acts as an obstacle that must never exceed the option value. Formally, the system

$$V \geq g, \qquad -\mathcal{L}V \geq 0, \qquad (V-g)(-\mathcal{L}V) = 0$$

expresses the complementarity of the two alternatives. In the continuation set where $V > g$, the product condition forces $\mathcal{L}V = 0$. In the exercise set where $V = g$, the differential inequality need not be an equality. This latter region cannot generally be specified in advance, so the problem is a free-boundary problem rather than a conventional linear PDE.

21.5 A lower bound for an American put's early-exercise region

Consider a non-dividend-paying stock, positive interest $r > 0$, and an American put with strike K. A very small stock price makes the immediate exercise reward $K - s$ close to K. Waiting until expiration gives no more than a payment bounded by K, but receiving K in the future has

discounted value at most $Ke^{-r(T-t)} < K$. For sufficiently small $s > 0$ and sufficient time remaining, immediate exercise can be superior to waiting. More precisely, the continuation value must be compared to $K - s$, and the optimal exercise boundary $s^*(t)$ delineates this region. This explains economically why the no-early-exercise proof for a non-dividend-paying call cannot be reused for a put: the cash benefit from immediate exercise goes in the opposite direction.

21.6 Penalty approximations and obstacle enforcement

For a discretized operator A and right-hand side b, a penalty method seeks a vector satisfying

$$Av - b - \rho(g - v)^+ = 0,$$

where $\rho > 0$ is large. If a component v_i falls below the exercise payoff g_i, the positive-part term becomes $\rho(g_i - v_i)$ and pushes the equation toward a higher value. If $v_i \geq g_i$, the penalty vanishes and the equation reduces to the continuation equation. The limit $\rho \to \infty$ encourages the hard obstacle constraint, but in finite precision a very large ρ may cause poor conditioning and make the numerical linear systems difficult to solve. A converged computed price must be checked for both complementarity residual and discretization error.

21.7 Why dividends can make early call exercise worthwhile

A call owner receives no dividends before exercise, while a stock owner does. With constant yield $q > 0$, a deeply in-the-money European call has approximate value

$$C^{\mathrm{E}}(s, t) \approx se^{-q\tau} - Ke^{-r\tau}.$$

Immediate exercise yields approximately $s - K$ when $s \gg K$. Their difference is

$$(s - K) - \left(se^{-q\tau} - Ke^{-r\tau}\right) = s(1 - e^{-q\tau}) - K(1 - e^{-r\tau}).$$

For sufficiently large s and positive q, the foregone dividend benefit $s(1 - e^{-q\tau})$ can dominate the financing benefit $K(1 - e^{-r\tau})$ of postponing strike payment. Hence the no-early-exercise argument for a non-dividend-paying call no longer applies. In a continuous-yield setting, the exercise boundary for an American call is therefore nontrivial, as with an American put.

Lecture 22

Numerical American Options and Portfolio Risk Optimization

22.1 Discrete linear complementarity systems

Let $g_i = g(s_i)$ be the discretized exercise payoff and v_i^n the option value at time-to-expiry grid level τ_n. For European options, an explicit step has the form $\widetilde{v}^{n+1} = (I - M)v^n$. For American exercise, the node value must not fall below the payoff. The resulting componentwise complementarity problem is

$$\begin{cases} v^{n+1} - (I - M)v^n \geq 0, \\ v^{n+1} - g \geq 0, \\ (v^{n+1} - g) \odot (v^{n+1} - (I - M)v^n) = 0, \end{cases}$$

where $\odot$ denotes coordinatewise multiplication. The explicit solution, when the step is well-posed, is the simple projection

$$\widetilde{v}^{n+1} = (I - M)v^n, \qquad v^{n+1} = \max\{\widetilde{v}^{n+1}, g\},$$

with the maximum taken componentwise. The projection is the discrete analogue of comparing continuation value with immediate exercise.

22.2 Implicit and Crank–Nicolson complementarity

Fully implicit and Crank–Nicolson schemes yield linear complementarity problems (LCPs) of the generic form

$$\begin{cases} Av^{n+1} - b^n \geq 0, \\ v^{n+1} - g \geq 0, \\ (v^{n+1} - g) \odot (Av^{n+1} - b^n) = 0. \end{cases}$$

For implicit Euler, $A = I + M$ and $b^n = v^n$ (subject to boundary-row adjustments). For Crank–Nicolson, $A = I + M/2$ and $b^n = (I - M/2)v^n$. Unlike European pricing, one cannot simply solve $Av^{n+1} = b^n$ everywhere because the exercise obstacle may bind at an unknown subset of nodes.

22.3 Projected iterations and penalty approximation

A projected solver alternates a linear-system update with a projection onto $v \geq g$, or uses a projected successive over-relaxation method. Another approach introduces a penalty parameter $\rho > 0$ and approximately solves

$$Av - b - \rho(g - v)^+ = 0$$

with the positive part applied componentwise. As ρ increases, violating the intrinsic-value constraint becomes increasingly expensive. Excessively large penalties may make the nonlinear system ill-conditioned, so both penalty error and iterative tolerance must be controlled. Newton-type active-set solvers exploit the fact that the unknown exercise region can be represented by a set of binding inequality constraints.

22.4 Markowitz mean–variance optimization

Let $R = (R_1, \dots, R_d)^\top$ be the vector of asset returns, $m = \mathbb{E}[R]$, $\Sigma = \mathrm{Cov}(R)$, and $x \in \mathbb{R}^d$ the portfolio weights. The return is $R_p = x^\top R$, with

$$\mathbb{E}[R_p] = x^\top m, \qquad \mathrm{Var}(R_p) = x^\top \Sigma x.$$

The classical minimum-variance problem under a target return m_0 is

$$\min_x \ \frac{1}{2} x^\top \Sigma x, \qquad \mathbf{1}^\top x = 1, \quad m^\top x = m_0.$$

If short selling is prohibited, add componentwise constraints $x \geq 0$. Assuming Σ is positive definite and moment constraints feasible, the unconstrained solution can be derived from Lagrange multipliers.

Proposition 22.1 (Minimum-variance weights without inequality constraints)**.** *Let $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, $B = \mathbf{1}^\top \Sigma^{-1} m$, $C = m^\top \Sigma^{-1} m$, and $D = AC - B^2 > 0$. Then the minimizing portfolio is*

$$x^* = \Sigma^{-1} \left(\frac{C - B m_0}{D} \mathbf{1} + \frac{A m_0 - B}{D} m \right).$$

Proof. Set $\mathscr{L}(x, \lambda, \gamma) = \frac{1}{2} x^\top \Sigma x - \lambda(\mathbf{1}^\top x - 1) - \gamma(m^\top x - m_0)$. Differentiating gives $\Sigma x - \lambda \mathbf{1} - \gamma m = 0$, so

$$x = \Sigma^{-1}(\lambda \mathbf{1} + \gamma m).$$

Enforce the two constraints:

$$\begin{pmatrix} A & B \\ B & C \end{pmatrix} \begin{pmatrix} \lambda \\ \gamma \end{pmatrix} = \begin{pmatrix} 1 \\ m_0 \end{pmatrix}.$$

Since the determinant is $D > 0$, matrix inversion yields $\lambda = (C - B m_0)/D$ and $\gamma = (A m_0 - B)/D$. Substituting proves the formula. □

22.5 VaR and Expected Shortfall optimization

When asset returns are non-Gaussian or portfolios contain nonlinear derivatives, variance may not adequately characterize downside risk. Let $L \in \mathbb{R}^d$ denote a vector of losses and $L_p = x^\top L$ the portfolio loss. One may minimize $\mathrm{VaR}_\alpha(L_p)$ or $\mathrm{ES}_\alpha(L_p)$ subject to budget, return, and exposure constraints. The VaR objective can be nonconvex, while Expected Shortfall admits a useful convex representation for suitable linear portfolio losses:

Theorem 22.2 (Expected Shortfall optimization identity)**.** *For an integrable loss Y and $\alpha \in (0, 1)$,*

$$\mathrm{ES}_\alpha(Y) = \inf_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{1 - \alpha} \mathbb{E}[(Y - \eta)^+] \right\}.$$

For simulated loss vectors $L^{(1)}, \dots, L^{(n)}$, this suggests the sample optimization

$$\min_{x,\eta} \left\{ \eta + \frac{1}{(1-\alpha)n} \sum_{i=1}^{n} (x^\top L^{(i)} - \eta)^+ \right\},$$

subject to the desired portfolio constraints. Introducing slack variables $z_i \geq x^\top L^{(i)} - \eta$ and $z_i \geq 0$ converts the objective to a linear program when the portfolio constraints are linear. This approach connects the course's simulation methods to computational risk management.

22.6 Complementarity as a discrete stopping decision

The explicit American-option update

$$v_i^{n+1} = \max\{g_i, ((I - M)v^n)_i\}$$

can be read as a one-step dynamic programming operation. The term $((I - M)v^n)_i$ approximates the discounted continuation value, and the intrinsic payoff g_i is the immediate exercise reward. For the update to be financially meaningful, the discretized continuation operator should preserve the relevant positivity/monotonicity properties. A negative finite-difference weight can cause a mathematically inconsistent continuation value and hence misclassify the exercise region. This is why monotone discretizations are particularly important for obstacle problems.

22.7 The convexity of sample Expected Shortfall

For a fixed simulated loss vector $L^{(i)}$, the function of portfolio weights x and tail threshold η given by

$$f_i(x, \eta) = (x^\top L^{(i)} - \eta)^+ = \max\{x^\top L^{(i)} - \eta, 0\}$$

is the maximum of two affine functions and is therefore convex. A nonnegative sum of convex functions is convex, so the sample ES objective

$$J(x, \eta) = \eta + \frac{1}{(1-\alpha)n} \sum_{i=1}^{n} f_i(x, \eta)$$

is jointly convex in (x, η). With affine budget/return constraints and convex position constraints, optimizing J is a convex program. The corresponding sample VaR objective is an order statistic of affine functions and generally lacks this convexity property. Convexity is a major computational advantage of ES for portfolio optimization.

Example 22.3 (Linear-program formulation)**.** Introduce auxiliary variables z_i for the positive deviations above η. Then sample ES minimization is equivalent to

$$\begin{aligned} \text{minimize} \quad & \eta + \frac{1}{(1-\alpha)n} \sum_{i=1}^{n} z_i \\ \text{subject to} \quad & z_i \geq x^\top L^{(i)} - \eta, \quad i = 1, \dots, n, \\ & z_i \geq 0, \quad i = 1, \dots, n, \\ & \mathbf{1}^\top x = 1, \qquad x \geq 0, \end{aligned}$$

plus any specified linear expected-return or exposure constraints. Every displayed objective and constraint is linear. Large simulation samples produce large but sparse linear programs, which standard optimization algorithms can handle without relying on a fragile numerical derivative of an empirical quantile.

22.8 Gaussian portfolio VaR and ES versus variance minimization

If $R \sim \mathcal{N}_d(m, \Sigma)$, the portfolio loss over the modeled horizon can be written $L_p = -x^\top R$ and is Gaussian with mean $-x^\top m$ and standard deviation $\sqrt{x^\top \Sigma x}$. Hence

$$\mathrm{VaR}_\alpha(L_p) = -x^\top m + z_\alpha \sqrt{x^\top \Sigma x},$$
$$\mathrm{ES}_\alpha(L_p) = -x^\top m + \frac{\varphi(z_\alpha)}{1-\alpha} \sqrt{x^\top \Sigma x}.$$

If the expected return is constrained to equal m_0, then $-x^\top m = -m_0$ is fixed. At conventional high confidence levels (so the positive normal coefficients apply), minimizing Gaussian VaR or ES is equivalent to minimizing the portfolio standard deviation. Both therefore select the same optimum as the Markowitz minimum-variance portfolio under identical linear constraints. This equivalence does not generally survive asymmetric, heavy-tailed, or option-induced return distributions. The goal of simulation-based risk optimization is precisely to accommodate such nonnormal behavior without imposing a multivariate Gaussian model.

Lecture A

Numerical and Probability Reference

A.1 Common distributions

The following identities are repeatedly useful in Monte Carlo implementations and stochastic models. For $Z \sim \mathcal{N}(0,1)$, $\mathbb{E}[Z] = 0$, $\mathbb{E}[Z^2] = 1$, $\mathbb{E}[Z^4] = 3$, and $\mathbb{E}[e^{tZ}] = e^{t^2/2}$. For $N \sim \text{Poisson}(\lambda)$, $\mathbb{P}(N = k) = e^{-\lambda}\lambda^k/k!$, $\mathbb{E}[N] = \text{Var}(N) = \lambda$. For $X \sim \text{Exp}(\lambda)$, $\mathbb{P}(X > t) = e^{-\lambda t}$, $\mathbb{E}[X] = 1/\lambda$, and $\text{Var}(X) = 1/\lambda^2$. If $X \sim \mathcal{N}(m, v)$, then $\mathbb{E}[e^X] = e^{m+v/2}$ and e^X is lognormal.

A.2 Black–Scholes Greeks for a non-dividend-paying call

With $\tau = T - t$, $d_1 = (\log(s/K) + (r + \sigma^2/2)\tau)/(\sigma\sqrt{\tau})$ and $d_2 = d_1 - \sigma\sqrt{\tau}$,

$$\begin{aligned} C &= s\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \\ \Delta &= C_s = \Phi(d_1), \\ \Gamma &= C_{ss} = \frac{\varphi(d_1)}{s\sigma\sqrt{\tau}}, \\ \text{Vega} &= C_\sigma = s\varphi(d_1)\sqrt{\tau}, \\ \Theta &= C_t = -\frac{s\sigma\varphi(d_1)}{2\sqrt{\tau}} - rKe^{-r\tau}\Phi(d_2). \end{aligned}$$

The derivations use $s\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$ and the chain rule. These Greek formulas provide independent test cases for finite-difference derivative calculations.

A.3 Geometric-average Asian option as a control

Suppose $0 < t_1 < \cdots < t_m = T$ and risk-neutral GBM. The logarithm of the discrete geometric average $G = (\prod_{j=1}^m S_{t_j})^{1/m}$ is

$$\log G = \log S_0 + (r - \tfrac{1}{2}\sigma^2)\bar{t} + \frac{\sigma}{m}\sum_{j=1}^m W_{t_j},$$

$$\bar{t} = \frac{1}{m}\sum_{j=1}^m t_j.$$

It is normally distributed, since linear combinations of jointly Gaussian Brownian values are Gaussian. Its variance is

$$\sigma_G^2 = \frac{\sigma^2}{m^2}\sum_{j=1}^m\sum_{k=1}^m \min(t_j, t_k).$$

Let $m_G = \log S_0 + (r - \sigma^2/2)\bar{t}$ and $v_G = \sigma_G^2$. If $v_G > 0$, its geometric Asian call value is

$$\begin{aligned} C_G &= e^{-rT}\mathbb{E}[(G-K)^+] \\ &= e^{-rT}\left[e^{m_G+v_G/2}\Phi(\widetilde{d}_1) - K\Phi(\widetilde{d}_2)\right], \\ \widetilde{d}_1 &= \frac{m_G - \log K + v_G}{\sqrt{v_G}}, \qquad \widetilde{d}_2 = \widetilde{d}_1 - \sqrt{v_G}. \end{aligned}$$

This explicitly known quantity is a strong candidate for a control variate when estimating arithmetic Asian calls.

Lecture B

Additional Simulation Techniques from the Instructor Notes

B.1 Common random numbers

Suppose two policies have output random variables $Y_A = g_A(U)$ and $Y_B = g_B(U)$ and the goal is to estimate $\theta_A - \theta_B$. Independent simulation gives variance $(\mathrm{Var}(Y_A) + \mathrm{Var}(Y_B))/n$ for a difference of two sample means. If the same uniforms U_i drive both alternatives, the paired estimate $n^{-1}\sum_i (Y_{A,i} - Y_{B,i})$ has variance

$$\frac{1}{n}\left(\mathrm{Var}(Y_A) + \mathrm{Var}(Y_B) - 2\,\mathrm{Cov}(Y_A, Y_B)\right).$$

Thus common random numbers help when the two outputs are positively correlated. This is especially valuable when comparing close option prices, hedge strategies, or alternative parameter values rather than estimating each independently.

B.2 Brownian bridge sampling

A Brownian bridge generates intermediate values conditional on already sampled endpoints. Given $0 \le s < t < u$ and Brownian values $W_s = x$ and $W_u = y$, the conditional distribution of W_t is normal with

$$\mathbb{E}[W_t \mid W_s = x, W_u = y] = \frac{u-t}{u-s}x + \frac{t-s}{u-s}y,$$
$$\mathrm{Var}(W_t \mid W_s, W_u) = \frac{(t-s)(u-t)}{u-s}.$$

Therefore, after simulating $W_T = \sqrt{T}Z_1$, one can sample $W_{T/2}$ conditional on (W_0, W_T), then fill in remaining midpoints recursively. This is an exact method for constructing a discretely sampled Brownian path. In quasi-Monte Carlo, allocating early low-discrepancy coordinates to the large-scale bridge components may reduce effective dimension and improve the quality of path-dependent option integration.